

**Box 19.2 Treadmill introduction**

- Walk the dog into the room and around the treadmill while it is off
- Walk the dog up onto the treadmill while it is off, giving treats while on the treadmill. Practice walking on at the back and off at the front. You may want to start to introduce commands for getting on the treadmill and getting off so that the sessions are controlled
- Have the dog sit by the treadmill when it is on, giving cookies so that he or she remains calm and becomes accustomed to the noise
- Place whatever device you will be using while doing the treadmill on your dog (harness/ safety vest/leash) and have them get back onto the treadmill
- Never tie the dog to the treadmill. Always hold the leash and stand next to the treadmill in case of emergency
- Slowly turn the treadmill on while feeding cookies to make it a positive experience. Remember that in the beginning taking one step should be rewarded. As time passes, reward for longer sessions
- You may need to hold the harness so the dog feels comfortable or show a favorite toy to motivate him or her to walk forward
- Unless you know if your dog can chew and walk at the same time (which many cannot) you should reward with praise and pats until the treadmill stops
- Increasing speed and time is not crucial until the dog is comfortable with being on the treadmill. Increasing speed can cause the dog to misstep and cause injury. Increased time can cause fatigue and muscle soreness and therapy setback
- The whole point is to have the dog walking on the treadmill in order to strengthen muscles, increase endurance, as well as a multitude of other reasons. However, the most important thing to remember is – make it fun!



**Figure 19.4** The land treadmill can also be used for advanced exercises as shown here. The dog has front feet on the ball and rear feet on the treadmill.



**Figure 19.5** Land treadmills can be used with feline patients. Dr. Sheilah Robertson's cat, Calvin, is on the land treadmill.

## Control, Supportive, and Assistive Devices

Harnesses, safety vests, and leashes should be used to control the patient during exercise. Harnesses that do not restrict shoulder motion are used for rehabilitation. For patients that are not ambulatory or ataxic, front and rear end harnesses are used to assist both the patient and the therapist. Booties are used if the surfaces are slippery or if the dog is weak and dragging of the feet may be a problem. Doing exercises over balls, or foam rollers, over peanuts or while in a sling can be very useful for neurological patients (see Chapter 10 for more details about this topic).

## Walking

Controlled leash walking is an excellent therapeutic exercise. It is easy to do and dogs usually enjoy it. Health benefits should include increased endurance, strength, cardiovascular fitness, and good mental stimulation. Unless the dog is confined for safety and only allowed outside for bathroom privileges, most dogs can start walking soon after surgery. Leash walks generally consist of 5 minutes of activity 2–3 times daily at first and progress each week by 1–5 minutes per walk depending on the patient. Be sure to specify the relative speed of the walk, and the need for continuous walking rather than the stop and sniff of a leisure walk. During inclement weather or cold winters, land treadmill walking may be preferable to walking outdoors. Treadmill walking, although it provides exercise, weight bearing, and strengthening, does not give the dog (or client) any mental stimulation (Saunders, 2007).

## Exercise Categories

In the following sections various exercise categories are discussed:

- Balance and proprioceptive exercises
- Core strengthening

- Hindlimb exercises
- Forelimb exercises
- Neurological rehabilitation exercises
- Exercises for cats.

## Balance and Proprioceptive Exercises

### Cavaletti Course

This helps with proprioception, strengthening, and gait retraining. Pole heights can vary and the distance is case-dependent. The poles are evenly placed apart and the animal walks over the poles very slowly. The slow pace allows the animal to step over the poles one paw at a time and recognize their paw location. The goal is to walk over the poles without touching them. It is best to start with 5 minutes twice daily. The pace can be progressed to a trot as patient strength increases, and distance between poles can be lengthened to encourage more limb extension in protraction (longer stride length).

### Balance Boards or Balance Discs

These are used for balance, proprioception, and strengthening legs. Place the targeted legs on the board, elevate the other limbs to the same height to equalize weight bearing, or elevate the other limbs above the board level to increase weight bearing through the affected limb(s). Control the patient with a harness. The board can initially be stabilized to allow only a small ROM in rocking as the patient learns to balance, then as the patient becomes more competent the board can be rocked back and forth. Balance discs or BOSU balls can also be used for this, and at home a firm cushion or air mattress. To increase the balance challenge, have the patient take treats from different locations on the board so they will have to shift position while maintaining balance. Start with a small amount of time twice daily and progress as the patient strengthens. Determine amount of time based on in-clinic observations. Watch out for fatigue and postural compensation as a guideline for when to stop, and give the client a time limit for home exercise, as well as red flags to look out for.

### Weave Cones

Weave cones are generally used to improve proprioception and increase core strength. The exercise can also pattern-improved for weight bearing when a limb is being protected after long-term adaptations to pain. Objects, normally 6–8 of them, are lined up about 1–3 feet (30–90 cm) apart (depending on patient size) and the dog is weaved in and out of the objects. Work to have an even number of bends each direction. The dog needs to learn to pivot sharply at the end of the pattern, and the therapist or client can use their body as a guide to prevent excess lateral motion and to keep the turns tight. Normally, this exercise lasts about 5 minutes twice a day but base the amount of time on observations in-clinic, looking closely for signs of fatigue (reduced control in the tight turns, reluctance to bend, slowing down, etc.).

### Figure Eights

This exercise is also performed using cones and is used to increase balance, coordination, spinal ROM, and for weight shifting from one limb to another. It is completed by walking the outline of the number “8” around two cones. The length of the figure eight normally is twice the height of the dog and the activity should be performed at a slow speed. The slow speed allows for an increase in spinal ROM. This should be done a few minutes at a time to avoid dizziness.

### Blocks

These are used in developing proprioception and core muscles. The animal places one paw on each block in a standing position and holds the stance for increasing periods of time. The blocks can be moved closer to mid-line or both paws can be on one block.

### Balance Beam

This is a length of plywood elevated as described above (in the section on Planks/Blocks/Stairs) that the dog walks along and sits on while maintaining their balance or posture. The goal is to have the dog walk along and not step off the beam.



**Figure 19.6** Uneven cushions: This patient is a recovering neurological patient who is working on balance and proprioception with the uneven surface. Core strength is also challenged in this exercise.

### Trampoline, Cushions, Air Mattress

These uneven surfaces help improve weight bearing as well as proprioception. The exercise can be done in many ways. Start with two paws on the uneven surface and move the patient around it, then have the patient walk across the surface (Figure 19.6). Progress to standing on the surface while the therapist applies gentle pressure to each hip or shoulder in a back and forth motion to challenge balance. Be sure to use a harness to control the patient so they do not lose balance and fall.

### Rhythmic Stabilization and Weight Shifting

Weight shifting is used to increase weight bearing and balance. Weight shifting should be done while the dog is standing. Place your thumbs over the dog's pelvis bones and your hands down their sides. Slowly sway side-to-side or front-to-back, making sure both legs are weight bearing. Do not use enough force



**Figure 19.7** Ball weight shift: This patient is recovering from a fibrocartilaginous embolism and is working on regaining strength in the rear legs. The therapist rolls the ball back and forth.

to cause the dog to lose their balance. This can also be performed in a supported stance, on a therapy roll, etc.

### Ball Work

This targets proprioception, allows for advanced strengthening and balance, and increases core and trunk stability. Most exercises can be performed on a Theraball™ (Figure 19.7). Once the dog has mastered exercises on solid ground, balls can slowly be introduced. To work on balance and proprioception, a large, egg-shaped ball should be used. The goal is to have the dog balance on the unstable surface and eventually to perform transitions such as sit-to-stand exercises. More ball work is described in other sections.

## Core Strengthening

### Crawling

This exercise is great for core strengthening, improving spinal mobility, ROM, and limb strengthening. It can be done anywhere and with anything; dogs can crawl under chairs, beds, boxes, agility tunnels, or they can just

learn to crawl along the floor. The higher the “crawling tunnel” that is created, the easier the exercise is for the patient. Begin with short distances and put treats along the tunnel to encourage the patient to move forward. Gradually increase the distance. Remember to add a command word and always reward while they are learning.

### Sit Up and Beg

This helps core strengthening, and hind leg strength. Initially have the dog sit squarely and then get up onto their back legs as though they were begging. Treats will help make this easier to teach. The dog may be unstable and the therapist may need to hold one front leg until the dog is stronger (higher is easier for the dog). Once the dog can hold the beg position, asking the dog to stand on their hind legs will strengthen the pelvic limbs. This is done with holding a treat higher. The dog is then asked to get back into the beg position. The up and down from beg to stand is like performing squats for humans. Eventually work up to two sets of 8–10 twice daily. This exercise should not be done if the dog has significant spinal or hip issues.

### Diagonal Leg Lifts

This exercise helps with core strengthening but also balance, weight shifting, and leg strengthening. The stance is achieved by lifting one leg off the ground along with the diagonal leg. Both legs are to be lifted at the same time and minimal support to the limbs should be provided so the dog must balance (Figure 19.8). Hold this pose for 10–30 seconds and repeat 8–10 times. When this exercise is easy for the dog, progress to having the dog stand on four blocks and then lift the diagonal legs. Finally add some weight shifting to cause the dog to balance further.

Diagonal leg lifts on a therapy ball are another possibility and these can be done as further challenge for the dog.

### Planking on a Ball or BOSU

There are many ways this exercise can be done depending on the focus of the exercise. The



**Figure 19.8** Diagonal leg lifts: This is a core strengthening exercise. As the patient's balance and core muscles improve the legs can be lifted higher.

dog may stand with their front legs on one peanut ball, their back legs on another ball, or front legs on a BOSU or ball and back legs on a balance disc or on the ground. The ball may also be moved while the dog is on it, rolled forward, or side to side while the dog balances. Alternatively the dog may be asked to keep front legs on the ball and walk around the ball.

#### Side Sit-Up

This exercise builds core strength but also increases spinal ROM. The dog should lie laterally. Using a treat, lure the dog up sideways, as if they were doing a sit-up sideways. Try to hold for 5 seconds then return to lateral. Do up to 10 reps per set. Dogs with spinal problems should be cleared before doing this exercise.

### Hindlimb Exercises

#### Sit-to-Stand

This exercise aims to improve strength in the pelvic limbs. When performing this exercise, it is important to assure that the dog sits

squarely and that on transition to stand, both pelvic limbs are used to propel the rear into extension. The patient should not pull themselves forward with the front limbs. A square sit occurs when the hips, stifles, and tarsi are in a straight line with the shoulders. The shoulders are perfectly aligned with the carpal joints. If the dog kicks one leg out to the side the therapist should have the dog stand again and sit while the therapist places their leg against the dog's paw, thus ensuring a square sit. Alternately, the dog could be asked to sit against the wall with the affected leg against the wall, thus preventing the leg from moving laterally. This exercise can be repeated over and over for increased strengthening.

#### Backward Walking

This exercise aims to strengthen the pelvic limbs and increase balance, coordination, and proprioception. We do not know the effects on specific muscles, only that the stance and swing phases of all four limbs are shorter with backward walking but that hip, stifle, shoulder, and elbow ROM are similar for forward and backward motion. There is some reduced ROM in carpus and tarsus, although this varied between the dogs studied (Vilensky and Cook, 2000). This exercise is easier to start with the dog parallel to a wall, a sofa, or anything straight. The therapist should hold a treat at their chest level and walk toward the dog saying "Back." Most dogs will try to turn while they are learning to walk backward, so it is important to give a reward when the dog takes a step backward. Another way of training this is to have a couch parallel to the wall just far enough from the wall that the dog cannot turn around. Lure the dog in forward with a treat, then say "Back" and encourage the dog to walk backward for another treat. This is an important exercise to teach all dogs and puppies as at some time in their lives. All dogs need good hindlimb awareness.

#### Side Stepping

This exercise aims to improve both forelimb and hindlimb strength as well as proprioception and balance. The therapist stands facing the

side of the dog, grasps the dog's collar in one hand, and holds the other hand on the dog's opposite hip. The therapist then walks toward the dog, encouraging the dog to side step. An alternate method is to use a treat to keep the head focused forward, one hand on the hip and gently walk into the dog. If there is a problem controlling the back end of the dog during this exercise, a front and back end harness can be used for extra back end control. This exercise needs to be performed in both directions to target the muscles symmetrically.

### **Incline Walking**

Incline walking helps build muscle in the rear legs due to the weight being shifted toward the back. On a treadmill, an incline has been shown to predominantly increase hamstring muscle activity versus gluteal and quadriceps activity (Lauer *et al.*, 2009). Incline walking increases stifle joint flexion (Holler *et al.*, 2010). This exercise should be started with shallow inclines and then steeper inclines can be added when the pet is comfortable and adjusting well. Serpentine or cross-hill walking can also be incorporated, where the patient walks in a zig-zag pattern across the hill up and down (see section on Principles of Exercise Program). Start with 5 minutes and gradually work up to 20 minutes twice daily, incorporating hills.

### **Stairs**

Stair ascent increases extension of the hip (coxo-femoral) and hock joints and results in reduced stifle extension when compared with level walking. The stairs also increase maximal flexion of the stifle and hock with each stride (Durant *et al.*, 2011). An example of therapeutic use would be for a patient with reduced flexion ROM in the hock (e.g., post-surgical osteochondritis dissecans) therapy would include stair ascent starting with low rise steps at the appropriate time in recovery. Descending stairs increases ROM in the hip, stifle and tarsal joints versus walking on a decline slope (Millard *et al.*, 2010).

### **Loving on Stairs/Couch**

This exercise is used to improve rear leg weight bearing and strength, ROM, and hip

extension. The client sits on the stairs/couch. With the dog's back legs firmly on the ground, lift up the front legs and place them one/two stairs up or on the couch cushion (height is dependent on dog's body length). This causes the dog's weight to be shifted to the pelvic limbs, and the weaker back legs must support most of the body's weight. While the pet is standing up on their back legs, praise them and give love or treats so that the time passes quickly.

### **Crawling or Tunnels**

This exercise has been described in the section on Core Exercises.

### **Cavaletti**

Cavaletti walking results in increasing flexion of the stifle and tarsal joints and increases extension of the stifle joint (Holler *et al.*, 2010). This exercise has been described in the section on Balance Exercises.

### **Zink-Zeus Get Up**

This exercise was created by Dr. Christine Zink as an exercise she used with her patient Zeus. It is one of the most challenging hindlimb exercises and one that many dogs dislike performing. With the dog lying in lateral recumbency, the therapist holds the top paw. While holding the paw so it cannot be used to assist the movement, a treat is used to lure the dog into a standing position. If the dog will not lay laterally, the exercise can be started with the dog in sternal recumbency. Initially most dogs can only do 1 to 2 reps of this exercise (Zink, 2013).

### **Ladder Walking**

This exercise aims to strengthen the legs and improve ROM, balance, coordination, and proprioception. The exercise is performed with a ladder placed flat on the ground. This exercise is like cavalettis but can change in difficulty. Initially the dog walks forward over the rungs but can be taught to walk backward and sidestep through the ladder. This exercise should start at 5 minutes twice daily and work up in frequency.

## Front Limb Exercises

### High Five Salute

High five salute helps with ROM and strengthening of front legs. To perform this exercise, the dog should be sitting. The dog then brings the paw up to their head and away as though saluting. This exercise does take some training for most dogs. When “giving a paw,” dogs usually only move their forelimb distal to the elbow. In this exercise, they need to move their shoulder, which allows the paw to get up to their head. In this way, the extensors of the shoulder and limb protractor muscles are targeted, and if the paw is placed in various locations then abductors and adductors can also be strengthened. Start by holding a treat in the hand. When the dog paws at the treat, the reward is given. Move the hand higher and into the position you wish the dog to stretch.

### Wheelbarrow

Wheelbarrow targets the front limbs and the core. This may seem like an easy exercise but it is quite challenging and needs to be done slowly and cautiously. First, never just pick the dog or cat up by their pelvic limbs as that is a great deal of load on them and you need to assess patient compliance. Start with lifting the patient by the caudal part of the

abdomen or under the pelvis so that the rear legs are a few inches off the ground. Observe contractions of the core muscles. Keep the stationary position for a while until the dog is comfortable with being lifted. Second, slow movement forward and backward can be added, making sure that the animal’s spinal alignment is monitored. Advancing the exercise, the rear legs can be lifted and the patient balances on the front limbs (Figure 19.9). Use treats to lure the patient. The spine should be perfectly straight to prevent injury. Do not push the patient too quickly; this can cause tripping, leading to the dog’s head hitting the ground. Do not perform this exercise with a patient with significant spinal issues, shoulder pain, or carpal hyperextension.

### Stairs

This exercise (as discussed earlier) is like walking on a decline and the stairs or decline should start out shallow and gradually become steeper as the patient progresses. Changes should only be added when the pet is comfortable and adjusting well. Time may be increased as the patient becomes stronger.

### Crawling or Tunnels

This exercise has been described in the section on Core Exercises.



**Figure 19.9** Wheelbarrow: Care should be taken when holding a dog for this exercise.

**Cavaletti**

Forelimbs show increased flexion of the carpus and elbow over cavalettis (Holler *et al.*, 2010). This exercise is described in the section on Balance Exercises.

**Play Bow**

This exercise aims to increase forelimb strength and flexibility and to promote core strengthening. The therapist holds a treat in one hand and the other hand is under the abdomen. The treat is moved to the floor and the dog's head and shoulders follow while the abdomen is being held up so back legs are standing and front legs are down.

**Digging**

This exercise aims to strengthen the front limbs, improve ROM in flexion, improve proprioception, and increase core strength. When performing this exercise, it is beneficial to have a command and a designated area where digging can occur. Sand, soft soil, or snow when in season are good substrates when they are not too packed down. Stones can hurt their paws and should not be used. A treat or favorite bone can be buried and dug up. Caution: some clients do not wish to encourage digging!

**Ladder Walking**

This has been described in the section on Hindlimb Exercise but also works well for forelimbs.

## Neurological Rehabilitation Exercises

See also Chapter 11 for more information.

**Assisted Standing**

This exercise is important to build and maintain muscles needed for balance, proprioception, and locomotion. These muscles become atrophied if not used frequently, as for example, in a paretic patient. A therapy ball, rolled towel, cushion, foam roller, or other device (depending on the size of the dog) is placed under the dog's abdomen. While performing

assisted standing, it is crucial to keep the pet's feet in normal anatomically correct standing position without knuckling. Larger dogs normally require additional abdominal support from a sling or back end harness. Frequently, the feet need to be positioned by two people. Smaller dogs and cats generally require only one person if their abdomen is supported by a Theraball™. Aim to keep the patient's spine as close to a normal standing posture as possible, not kyphotic. Assisted standing should only last until the first signs of fatigue. After a brief rest, it can be repeated, but do not exhaust the patient.

**Physioball Work**

Balls are particularly important for rehabilitation of neurological patients (Figure 19.10). They can be used for assisted standing as described earlier, as well as for assisted walking and to regain balance and coordination. Frequently, the front end of the patient is placed on the ball and the ball is rocked back and forth. Foot placing by a second therapist may be required.

**Weight Shifting**

Neurological patients need to practice balancing on three limbs to mimic the transient



**Figure 19.10** Sit-to-stand on the ball may need two people in the case of a large dog.

unloading of a limb during ambulation. To perform this exercise, the therapist places the dog or cat in standing position and lifts one limb. When they start to sway, the limb is replaced. All four paws are rotated through this exercise.

### **Treadmill Gait Training**

Land treadmills are used for gait retraining for neurological patients. Many times, the patient must be supported with a sling. The therapist moves the dog's legs in walking gait pattern: RR to RE, LR to LF (Zink, 2013). Skin sensation stimulus (vibrations, scratching, or even e-stim) can be used to overcome spasticity. As the gait becomes more normal, incline can be added.

### **Proprioceptive Neuromuscular Facilitation Patterns**

Proprioceptive neuromuscular facilitation (PNF) has been around since the 1930s when neurology physician Herman Kabat began to use the technique on his patients. He found that by stimulating the distal segments, the proprioceptive nerves in more proximal segments became stimulated. His purpose was to enhance and create movement in areas where the neurological system has been compromised. His techniques were based on the principles that describe the rhythmic and reflexive actions that lead to coordinated motion. PNF uses the body's proprioceptive system to facilitate or inhibit muscle contraction (Stillman, 1966). PNF patterns mimic the dog's running motions and other normal functions of daily living (scratching, digging). To perform a PNF pattern for running, lay the dog on their side and mimic the running pattern, then mimic in a supported stance if possible. The therapist should use one hand to mimic the ground contact at the appropriate part of the gait cycle (Edge-Hughes, 2012).

### **Tactile Stimulation**

Tapping or using a vibrating massager over a muscle belly aims to elicit muscle contractions and stimulate the neural receptors

(muscle spindle cells and Golgi tendon organs) in muscle and tendon. This should be done for 3–5 minutes a few times a day as part of the nursing care for animals recovering from paralysis. Brushing, scratching, pinching, or using a vibrating massager provides additional sensory stimulation to skin receptors to increase input into the nervous system.

### **Tensor Bandaging, Thundershirts™, or Snugglis™**

The principle of using tensor bandages or Thundershirts for neurological dogs is to increase sensation in cutaneous and deeper tissues and so increase body awareness. The increased sensation input to higher centers can help with neurological coordination.

### **Exercises for Cats**

Cats can be challenging when it comes to rehabilitation. The principles of exercise are the same for cats as for dogs but the personality of the cat may limit what the therapist can do. Cats are commonly presented for rehabilitation due to chronic pain from injuries or because of osteoarthritis, which is common in cats. Cats can require rehabilitation following surgery for such conditions as cruciate rupture or patellar luxation. A few examples are listed here.

### **Proprioceptive Exercises**

These may include balance and rocker boards or standing on a therapy ball or using a rocking chair or glider as the uneven surface. Initially, some cats will only tolerate 30 seconds of exercise but may work up to 2 or 3 minutes as they become more comfortable. It is important that the therapist understand cat body language and discontinue the exercise as soon as the cat starts to become agitated.

### **Treadmill Training**

Cats can be trained to walk on land or underwater treadmills. If the cat is amenable to wearing a harness, then leash walking and use

of a land treadmill with assistance can be used to improve weight bearing and limb function (Millis *et al.*, 2014). Movement can be encouraged by attaching a feather or small toy to a string and pulling it in front of the cat.

### Cavaletti Rails

Some cats can be trained to walk over cavaletti rails, particularly if they are food motivated. Some cats will respond to treats – others need to be encouraged by toys or by chasing a laser pointer over the rails.

## Case Studies

### Case Study 1: Iliopsoas Injury and Partial Cranial Cruciate Ligament Tear

#### Signalment/History

A 6-year-old male neutered Golden Retriever presented for rehabilitation 1 week after tentative diagnosis of a partial cranial cruciate tear and an iliopsoas muscle injury. Radiographs and musculoskeletal ultrasound were performed to confirm diagnosis. The dog had been injured by falling from an A-frame during an agility run 3 weeks prior to presentation.

#### Clinical Signs

Iliopsoas pain and spasm on right side. Slight drawer in flexion right stifle with very mild restricted ROM in flexion. Partial weight bearing and mild muscle atrophy right pelvic limb (1 cm discrepancy in circumference). Pain on ambulation, particularly extension of right pelvic limb and in left sacroiliac joint. Shoulder muscles high resting tone, myofascial trigger points in triceps.

#### Goals

Relieve pain, normalize ROM, improve ambulation, strengthen muscle, improve weight bearing.

#### Therapy

Therapeutic ultrasound, passive range of motion (PROM), and stretching helped

relieve pain. Pain medication and herbals were also prescribed. Hydrotherapy was used to aid with strengthening. Therapeutic exercises in hospital and at home were used to aid recovery.

#### Exercise Plan

Warm-up right pelvic limb with hot packs applied for 10 minutes to iliopsoas and biceps femoris muscle. Massage and stretching was done before exercise to improve circulation and help with pain relief. Exercise sessions up to 20 minutes. Icing was done post exercise for 5–10 minutes. In hospital, exercises consisted of cavaletti course (cones with poles 4 inches (10 cm) high) for 5 minutes total with a few 10- to 30-second breaks. Weave poles 30 inches (76 cm) apart 6 cones in total; figure eights, slow circles, and turns for 5 minutes total with a few 10- to 30-second breaks; rhythmic stabilization on a rocker board 3 sets of 30 seconds each; side stepping initially 3 sets of 30 seconds each; 5 sit-to-stand exercises; backward walking 8–10 feet (2.4–3 m) 3 times; and treat stretches and twists. In hospital, exercises were to be done 2–3 times weekly.

#### Home Exercise Plan

Client was instructed to hot-pack the pelvic limbs for 10 minutes before exercising and ice for 10 minutes following exercise. After hot-packing the limbs, stretching was performed with the dog standing and one leg stabilized above the stifle, mimicking sitting position with a stretch and hold of 15 seconds. After that the leg was moved in a bicycling motion. Lastly, the limb was stretched gently in adduction and abduction. Stretches were performed 5 times before the exercise session.

#### Home Exercises

- Loving on the stairs with a twist – This patient had pressure added to each side of the body to sway him back and forth while he was standing on a cushion. This increased the firing of the muscle that allowed him to improve overall balance

even more, since he was standing on an uneven surface. This was to be performed twice daily for 10 repetitions.

- Cavalettis, weave poles and figure eights were to be done twice daily and at a slow pace.
- Challenged standing—This exercise was performed at least twice daily with 10–15 repetitions per performance.
- Play bow—This exercise was done twice daily and worked up to 15 repetitions each session.
- Side stepping—This dog had a lot of obedience training and already knew how to walk sideways, so the duration was increased to 20–30 feet (6–9m) 5 times twice daily.
- Sit-to-stands—This dog sat on their left hip and popped their right hip out. To correct this the client needed to position the dog against a fence/wall with their right leg touching the wall while he sat. This removed the ability for the hip to pop out and the patient sat. Repetition was set at 15 sit-to-stands 3 times a day every day and the emphasis was on square sitting.
- Treat stretches or treat twists—Twice daily for 5 minutes combined.
- Backward walking and backward walking upstairs/hill—In this case he was already able to walk backward, and was trained to do so previously, so the length was increased to 20 feet (6m) 3 times a day with 5 repetitions each session.

Three weeks after the initial rehabilitation consult the client was having a few issues with the at-home exercises. The client was also finding it hard to keep the dog rested, as there was more than one dog in the house.

- Walking backward—There was a complication with the backward walking as the dog started to turn while walking backward since he was still favoring the right rear leg. It was recommended to walk along the fence line instead so that the dog was unable to turn.
- Walking backward up hills—The dog would only do two stairs backward then

stop. The easiest solution would be to continue doing the two stairs multiple times per day if the dog was not willing to walk backward up more than those two steps.

- Leash walks were allowed for 10 minutes 2–3 times daily. The duration could be increased by 10 minutes each week but no off-leash running/play was allowed.

Five weeks after the rehabilitation consult the dog was doing much better. The iliopsoas muscle did not have any trigger points and muscle mass started to rebuild in hind right leg. The dog was sitting squarely almost every sit and the restriction of leashed bathroom break was removed.

Home exercises were continued with the addition of 30-minute walks and zigzag hill walking.

### **Outcome**

By 10 weeks the dog had normal ROM and even muscle girth with no visible lameness and was discharged from rehabilitation.

## **Case Study 2: Intervertebral Disc Disease Conservative Management**

### **Signalment/History**

A 13-year-old F/S Bichon Frise presented 7 days after acute incident of paralysis. Radiographs were not presented. Lesion localized on examination to T3–L3. Normal urination and defecation. Previously treated with methylprednisilone and on oral prednisone at presentation.

### **Clinical Signs**

Paresis of both pelvic limbs. Unable to walk or weight-bear on hindlimbs independently. Pain score 3/10. Panniculus reflex stopped at T11, localizing lesion to T12–T13 area. Superficial and deep pain sensations were present, tail wag present, withdraw present bilaterally. Left pelvic limb weaker than right, mild muscle atrophy of gluteals, hamstring group, and quadriceps. Shoulder muscles high tone with myofascial trigger points in triceps and trapezius muscles. Normal ROM all joints except

stifles which had decreased ROM with bilateral medial patellar luxation (grade 1). Magnetic resonance imaging and surgery were not an option for this patient. Client elected for non-surgical therapy with rehabilitation.

### Goals

Decrease pain, improve neurological function, strengthen muscle, and regain ability to ambulate properly.

### Therapy

Two weeks of rest was instituted before any exercise therapy could start. During that time the client brought the dog to the rehabilitation center for laser, acupuncture, and electroacupuncture. At-home PROM exercises were performed twice daily and then home application of neuromuscular electrical stimulation (NMES) to hamstring and quadriceps muscles bilaterally once daily for 10 minutes, followed by massage. Prednisone was continued although dosage was decreased. Gabapentin and herbal medications were added for pain. Traction socks were placed on both back feet to prevent excoriations. After 2 weeks, the patient began exercise therapy including underwater treadmill therapy as well as continuing acupuncture for 3 more weeks.

### Exercise Therapy

Initially, the dog needed assistance with ambulation as she was unable to sit or rise properly. The first exercises done (also taught to the client) were proper transition from sternal lying position to sit, and on to stand with tapping of the pelvic limb muscles to facilitate muscle firing during the transition. The left hind required more stimulation as it was weaker. PNF patterning for walking was taught to the client and was performed in-clinic. PNF patterning for scratching the ear was also demonstrated. Square sit-to-stands with assistance were performed along with assisted standing. Gait patterning was performed in the underwater treadmill.

Once the dog was stronger and more functional, unassisted sit-to-stands began,

and assisted walking on land and rocker board work were introduced. Now the patient began to walk and had a “disconnected” gait. A Snugli was put on the dog to help coordinate the front and back end. NMES was discontinued when the dog started ambulating on her own. Cross-legged standing and weave poles were added to the exercises. The client elected to continue underwater treadmill therapy and to do home exercises.

### Home Exercise Program

- Sit-to-stands 10 reps 3 times each.
- Loving on the couch – Hold for 30 seconds for 6–8 reps.
- Cavaletti course with pool noodles – 5 minutes twice daily.
- Treat stretches – 8–10 reps of each exercise.
- Walking up to 20 minutes 2–3 times daily and increased as dog became stronger.
- Hill walking 5 minutes as part of a 15- to 20-minute walk.
- Low steps were later added.

### Outcome

Therapy goals were achieved within 4 months.

## Conclusion

Land exercises are crucial to the physical rehabilitation program. The veterinary technician/nurse will be instrumental in carrying out the plans that their supervising veterinarian has designed. The rehabilitation nurse will be in communication with the client teaching many of the exercises discussed in this chapter. Most often, demonstrating the exercises to the client and asking them to demonstrate the exercises back to the nurse is the most effective way of teaching. In addition, written instructions and, if possible, images should be given to the client as aids. It could also be helpful to have the client video their pet while it is performing the exercises. This can be reviewed by the rehabilitation therapist and nurse. The exercise plan can be adjusted by the therapist as the patient progresses.

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20

Therapeutic Exercises Part 2: Hydrotherapy (Aquatic Therapy)

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Introduction

Principles of Hydrotherapy

The definitive goal of aquatic therapy is ultimately to make the patient more functional on land in a faster time frame in comparison to land-based therapy alone. Swimming and underwater treadmill (UWTM) walking remain the most commonly encountered aquatic therapies in veterinary medicine.

Physics of Hydrotherapy

To gain an understanding of how aquatic therapy differs from land-based therapy you must understand the properties of water and how those properties act on the patient during hydrotherapy.

Density and Buoyancy

The density of an object determines whether it will sink or float when placed in water. In hydrotherapy, we look at the relative density of an object, which is described as a ratio between the object’s weight and the weight of an equal volume of water. If an object’s density is less than that of water it tends to float and if an object’s density is greater than that of water the object tends to sink. The relative density of an object, or in our case a patient, is dependent on the patient’s body composition and it is related to the patient’s skeletal mass, muscle mass, and body fat percentage (Prydie and Hewitt, 2015). When a patient is immersed in water there are two main forces acting them – gravity and buoyancy. Gravity is attempting to submerge the patient, while buoyancy provides upwards thrust, which

counteracts gravity. A patient's center of buoyancy depends on their body conformation, and on the volume of fat and muscle that makes up their body. The water level of an UWTM can be changed as required, and the percentage bodyweight carried by a dog's limbs while using the treadmill can be changed by varying depth. Basically, the deeper the water, the less bodyweight is loaded onto the limbs (Tomlinson, 2013).

### **Hydrostatic Pressure**

Hydrostatic pressure is the sum of pressure on a body surface exerted by the fluid that it is immersed in (Connell and Monk, 2010). The hydrostatic pressure exerted on the patient is directly proportional to the depth and density of the water that they are standing in. Hydrostatic pressure increases with increased depth and density of water. Density of water is temperature-dependent. Increased hydrostatic pressures during hydrotherapy can reduce edema by moving tissue fluid from the extravascular space to the intravascular space. Increased hydrostatic pressure also has a direct effect on cardiac function; peripheral blood flow venous return has been shown to increase as hydrostatic pressure increases, and this can lead to a 10–15% drop in heart rate (Levine *et al.*, 2014). It has also been established that increased hydrostatic pressures during aquatic therapy can reduce pain during exercise. The elevated hydrostatic pressure of water blocks the pain response by acting on mechanoreceptors in the skin, which send signals to the cerebral cortex, downregulating pain pathways (Prydie and Hewitt, 2015).

### **Viscosity and Resistance**

Water molecules are attracted to each other and to objects moving through them. This attraction generates frictional resistance to movement of an object through water. Marine animals are adapted to have streamlined bodies that enable lower friction (drag) when moving through water. Our canine patients lack these adaptations and we can use this to our advantage; the drag of move-

ment through water provides more resistance to muscles than walking through air (Prydie and Hewitt, 2015). Viscosity is a measure of fluid friction, with more viscous substances generating more resistance to movement. We can alter the viscosity of the water in a pool or UWTM by altering the temperature of the water. Cold water is denser than warm water and is therefore more viscous and vice versa. The friction created by moving through water helps provide sensory feedback to patients via their skin. This may help, particularly in the rehabilitation of neurological patients who have damaged sensory positional awareness. Gait is exaggerated and gait speed is slowed by water viscosity. This is of benefit in patients recovering from neurological injury and is also useful as it allows more accurate assessment of a patient's gait, especially when walking on a UWTM.

### **Surface Tension**

The attractive force exerted between water molecules on the surface of a body of water is described as surface tension. Water molecules stick to each other with more force at the surface of a body of water than underneath the surface. This means that there is higher resistance to movement at the surface of a body of water in comparison to below the surface. For patients, this means that it requires more effort to move a limb in and out of water ("break the surface of the water") than if the limb remains submerged 100% of the time. As therapists, we can use this to increase force of motion on a joint (or limb segment) by adjusting where the surface of the water lies.

### **Refraction**

The path of light is altered as it passes across the boundary between two substances of differing density (e.g., air to water). This is known as refraction. It alters the patient's depth perception, which can make animals misjudge their foot placement on steps and ramps during water entry and exit. Care must be taken when transferring patients into and out of water. The refraction of light

may also alter the therapist's perception of a patient's limb movement under water.

## Evidence

There is a paucity of evidenced-based literature within the field of veterinary aquatic rehabilitation in comparison to human rehabilitation. There is currently a need for well-constructed research projects to evaluate the benefits of aquatic therapy across an array of orthopedic and neurological conditions. The main areas of research so far have revolved around using aquatic therapy to rehabilitate patients after cruciate ligament surgery and have largely been driven by the economic importance of cruciate disease (Wilke *et al.*, 2005).

In humans, underwater exercise has been shown to generate less cardiovascular and respiratory demand than equivalent land-based activities (Yoo *et al.*, 2014). This is of use in patients with coexisting disease or in obese patients who have low levels of athletic capability. Water temperature plays an important role in human aquatic therapy. Cold water temperatures have been shown to increase the energy expenditure of patients during aquatic therapy (Versey *et al.*, 2013), but the same evidence cannot be assumed to be directly transferable to veterinary aquatic therapy.

### Which Patients will Benefit?

There is substantial research available which indicates that muscle atrophy begins within 24–48 hours following immobilization. The muscles first affected are the postural muscles, which contain a large proportion of type I muscle fibers (Randall, 2010) (Box 20.1).

### The Orthopedic Patient

The most common orthopedic conditions treated with aquatic therapy are cranial cruciate disease, hip dysplasia, and osteoarthritis (Goldberg, 2015). Patients visiting your clinic for aquatic therapy will either present for non-surgical management of their orthopedic disease or for post-surgical rehabilitation.

#### Box 20.1 Aims of hydrotherapy

- Enhance preoperative fitness and conditioning (e.g., before a hip replacement)
- Complete resolution of clinical signs (e.g., after cruciate surgery)
- Postoperative restoration of some function (e.g., in cases of intervertebral disc disease)
- Merely palliative (including analgesia)
- Increase cardiovascular fitness
- As an adjunct to obesity management
- As a “fun” form of exercise

Source: Prankel (2008).

Non-surgical management of veterinary orthopedic disease compromises pharmaceutical management of pain, weight management, and rehabilitation therapy (Waining *et al.*, 2011). Patients with degenerative joint disease suffer from loss of muscle mass, reduced joint range of motion, and joint pain. These are all problems which can be effectively managed with aquatic therapy. Orthopedic disease of the canine coxofemoral joint is often managed with rehabilitation. Research has highlighted that between 70% and 90% of patients with hip dysplasia will have clinical signs attributed to the disease process and therefore most patients with hip dysplasia could benefit from aquatic therapy (Farrell *et al.*, 2007). Research has confirmed patients with degenerative joint disease benefit from hydrotherapy. Twice weekly hydrotherapy over an 8 week period has been proven to improve the function of joints affected by degenerative joint disease (Nganvongpanit *et al.*, 2014b).

Aquatic therapy is an integral component of the post-surgical rehabilitation of patients undergoing surgery for orthopedic disease. Research has shown that over US\$1.32 billion dollars a year are spent on the treatment of cranial cruciate disease in canines (Wilke *et al.*, 2005). Results of a study suggested that following surgical management of cruciate disease in dogs, swimming resulted in greater extension and flexion of the stifle and tarsal

joints than walking alone, which helps accelerate a return to normal function (Marsolais *et al.*, 2003).

### Neurological Patients

Intervertebral disc disease and degenerative myelopathy are the two most common neurological diseases treated using aquatic therapy in our practice. Aquatic therapy should be the cornerstone of rehabilitation in neurological patients. It can be tailored to accommodate patients with varying degrees of neurological disease from quadriplegia to mild muscular weakness. Standing in water is assisted by buoyancy, which provides a simple and comfortable means to maintain a standing position in non-ambulatory patients (Prydie and Hewitt, 2015). Standing can benefit lung inflation and cardiovascular fitness in a patient who has been non-ambulatory for some time. Pool- or UWTM-based aquatic therapy is also more likely to encourage limb movement earlier in the recovery period than similar therapeutic exercises carried out on dry land. The resistance provided by the water helps to strengthen muscles while the buoyancy of the water assists patients in reaching and maintaining a standing position with less exertion (Prydie and Hewitt, 2015).

Therapy in the UWTM may be more appropriate than a pool for neurological patients once they begin to develop a partially coordinated gait pattern, as the treadmill will provide a more controlled movement during which the gait pattern can be assisted to correct motion. The controlled motion provided by UWTM-based exercise places less strain and torque on the spine and encourages normal gait (the action of the treadmill belt drawing the limb caudally in stance phase along with the exaggerated flexion needed to bring the limb through swing phase in the water). Patients with neuromuscular disease are also likely to benefit more from UWTM-based therapy in comparison to free swimming, as the UWTM will help to reinforce a proper gait pattern and provide vital tactile feedback during footfall onto the

treadmill belt. Water level and treadmill speed can be adjusted to achieve an optimal gait pattern along with the ability to provide assisted active range of motion as needed. The UWTM can also allow the impaired patient to perform movement against gravity earlier during therapy than they would be able to during assisted exercise on a land treadmill.

Aquatic therapy also has several ancillary benefits for the immobile and weak neurological patient. Hydrostatic pressure and elevated water temperatures can help to ease muscle pain, reduce edema and promote lymphatic drainage (Prydie and Hewitt, 2015). When carrying out aquatic therapy on patients with neuromuscular disease it is important to bear in mind that these patients fatigue very rapidly and they should have their head and neck fully supported to prevent aspiration of water during therapy (you can use a pool float or an inflatable neck pillow). Swimming in a pool is often limited to very short periods of time (60–120 seconds) in these patients every 2–3 days, as this is the best way to avoid fatigue and exacerbation of weakness at home from this fatigue. It is best to have the patient wear a life vest (buoyancy aid) for their own safety. Hands-on support to the chest or abdominal areas can bring about a more comfortable swimming position, putting less strain on the spine.

### Athletes

There are a wide variety of sports in which canine athletes currently compete; a few examples include obedience, agility, herding, dock-diving, disc-dog and flyball, coursing, hunting and field trial, sledding, and skijoring. Working dogs such as those in police protection and detection work and search and rescue dogs also fall within the category of athletes. These sports (and jobs) require strength and aerobic fitness. Some activities require endurance. Aquatic therapy can provide vital post-injury strengthening and conditioning in rehabilitation of the canine athlete. Aquatic therapy can also play an integral role in training of canine athletes

(strength and endurance). It can be used in the off-season to maintain condition. Maintaining a level of fitness during the off-season helps to prevent injury when the patient returns to competition, though it should be emphasized that sports-specific training (for example jumping) is still needed.

Aquatic therapy engages and trains muscle groups that are difficult to recruit during land-based exercises, and therefore helps add to the overall fitness of the patient. Water-based therapy is less strenuous on the patient as it does not cause the same repetitive concussive forces on a patient that training on dry land does. An increase in joint flexion during aquatic therapy helps to increase a patient's active range of motion during land-based activities (Marsolais et al., 2003).

## Equipment

### Pool Design

Prefabricated pools offer an “off-the-shelf” option to centers wishing to introduce pool-based aquatic therapy. Prefabricated pools are manufactured in a variety of sizes and tend to command less space than a custom-built pool. They can be surface mounted or sunken to meet the design constraints of the facility. One key advantage over a custom-built pool is that prefabricated pools can be moved if the facility decides to move and they tend to have water management plant systems incorporated into the product (or plant systems that can be easily moved). Decking can be built around an above ground pool to provide ramp access from ground to pool level.

Custom pools provide a permanent solution to introducing aquatic therapy to a facility. The financial commitment associated with installing a custom-built pool means the facility must not be planning on moving in the medium- to long-term future. Custom pools give design flexibility in comparison to prefabricated pools, and they can be designed to fit into the available space. A custom pool



**Figure 20.1** Pool entry. *Source:* Courtesy of Wendy Davies.

also allows for the internal design of the pool to be tailored to the needs of rehabilitation unit. Different methods of entry and exit (i.e., ramp and steps) down into the water can be added (Figures 20.1 and 20.2). Working platforms of varying depths can also be added to provide areas on which to rest animals and provide static water-based therapy. Custom pools also add flexibility when it comes to water management plant design. If a custom-built pool becomes so busy the water management system is unable to maintain water quality, the systems can usually be easily upgraded.

Pool jets are an option that can be added at the time of pool installation. Underwater jets generate turbulence in the water that increases the amount of drag and consequently increases the amount of work being done by a patient (Box 20.2).

As pool size increases it should be kept in mind that specialist air-handling equipment may be required to manage the air quality in the therapy suite. Humidity levels and



**Figure 20.2** Pool entry side view. *Source:* Courtesy of Wendy Davies.

#### Box 20.2 Benefits of swimming

- Totally non-weight-bearing
- Maximum active range of motion of joints
- Non-ambulatory patients with paraparesis/paralysis
- Improved core and trunk strength
- Cardiopulmonary conditioning
- Endurance for cross-training
- Fun for patient
- Facilitates passive range of motion/all body work

*Source:* Chiquoine *et al.* (2013). Reproduced with permission of John Wiley & Sons.

temperature must be controlled to maintain patient and therapist comfort as well as prevent damage to the fabric of the building due to excessively high humidity levels.

There are large costs associated with heating a pool/UWTM every day. To conserve energy and control heating bills careful consideration should be made when planning

how to retain and maintain the temperature of the water in the pool/UWTM. Pools tend to circulate water 24 hours a day to maintain water quality, so ensuring all pipework is lagged with proprietary heat-retention products can help retain a lot of heat in the pool. In addition, it is recommended to use a pool cover to maintain the heat in the pool while it is not in use. The pool cover should be insulated with foam or a similar material to ensure as much heat as possible is retained. UWTM differ from pools in that the water used in the treadmill tends to be stored in a tank. Non-insulated tanks expend a lot of energy keeping the water to temperature even when the treadmill is not in use, so it is recommended to insulate the water storage tank to save energy.

#### Underwater Treadmill

There are several companies manufacturing UWTM across the globe. When looking to invest in an UWTM first explore all the options available to you in your geographical location. Second, aim to visit rehabilitation centers that are using the treadmills you are considering and ask the therapists using the machines to give you all the positives and negatives regarding that unit (Box 20.3).

When investing in a UWTM one should examine the minimum and maximum speed of the unit and what is the smallest incremental speed change that can be made on the unit during a therapy session. To effectively rehabilitate small patients after neurological injury you need to be able to alter belt speeds by small increments (0.1 mph) so small changes in belt speed can be made to tailor therapy to the exact needs of the patient. It should be noted that treadmill speeds are usually not calibrated to actual speed so if the facility has two UWTM, they may differ in actual speed. Before committing to a unit ensure you check whether speed can be altered during the therapy session. Does the machine need to be stopped, the belt speed adjusted, and then the belt restarted? The ability to add an incline during therapy is an

### Box 20.3 Benefits of the underwater treadmill

- Improved active range of motion compared to land
- Permits partial weight bearing
- Proprioceptive gait training
- Improved balance while walking
- Cautious fracture loading
- Builds lean muscle in limbs
- Helps timid/new swimmers get started
- Speeds gait retraining

Source: Chiquoine *et al.* (2013). Reproduced with permission of John Wiley & Sons.



**Figure 20.3** Dog swimming in pool. Source: Courtesy of Wendy Davies.

important addition that many treadmills offer. Some units automate adding inclination to the belt via the control unit, allowing you to change the angle of inclination during a therapy session (Figure 20.3 and 20.4). Most units available on the market require the inclination of the belt to be set manually prior to commencing the therapy session and



**Figure 20.4** Weight bearing is reduced by 15% if the water is filled to the level of a patient's stifle in an underwater treadmill. Source: Courtesy of Wendy Davies.

do not allow the inclination to be altered once the UWTM has had water added.

UWTMs are complicated mechanical machines that require daily maintenance in the clinic and regular maintenance from the manufacturer. Cleanliness is incredibly important to keep an UWTM running effectively and daily maintenance protocols should be established to maintain the machine.

### Ancillary Equipment

A buoyancy jacket, or life vest, helps a patient to float in the pool or in the UWTM if the treadmill is filled to the patient's shoulder blades. Buoyancy jackets allow the therapist to remain in control of patients via handles and straps. A jacket also helps the patient maintain a level stance in the water. An exception to the use of a buoyancy jacket to keep a patient level is the amputee. The rotational instability which is already a problem

for the amputee in the water can be made worse by increasing buoyancy (Box 20.4). Buoyancy jackets are very useful in patients who are fearful or lack confidence in the pool as they help the patient maintain their head well out of the water (Figure 20.5). The jackets must be carefully sized for the patient to ensure they do not restrict limb or neck movement.

Once a patient no longer requires a buoyancy jacket during therapy, use a non-flotation jacket or a robust harness during their therapy session. Harnesses or jackets provide the same patient control benefits as buoyancy jackets without providing flotation.

#### Box 20.4 Equipment used with therapeutic swimming

- Life vest
- Harnesses
- Slings
- Head wraps
- Balance equipment
- Leg weights
- Resistance swim mitts
- Pool noodles
- Toys
- Food

Source: Chiquoine *et al.* (2013). Reproduced with permission of John Wiley & Sons.

Patients with neurological conditions affecting limb strength may require additional flotation for therapy to be beneficial. Long foam cylinders, commonly called pool noodles, are useful for both pool and UWTM therapy and can be placed under the thorax or abdomen to provide additional flotation during treatment sessions. These foam cylinders make excellent bumpers for use in the UWTM to prevent patients from placing their limbs on the static side panels either side of the treadmill belt. An alternate bumper for this latter use is a fender float (Figure 20.6).

Motivation is key to ensuring a patient remains focused and attentive during therapy. Owners should be encouraged to bring along toys that the patient enjoys playing with. Toys can be used to encourage a patient to swim laps or lengths of a pool and can be used to promote consistent gaiting during UWTM therapy. Toys should be cleaned prior to use to minimize pool contamination and should be in good repair to prevent the patient possibly swallowing broken pieces. Toys should also easily fit in a patient's mouth; excessively large toys that require the patient to open their jaws to the maximal extent may put the patient at risk of aspiration of water into the respiratory tract.

There are various therapeutic aids that can be used to make aquatic therapy more

**Figure 20.5** Harness for controlling patient in underwater treadmill and keeping head above water.  
Source: Courtesy of Wendy Davies.





**Figure 20.6** Fender floats prevent patients from placing their limbs on the static side panels either side of the treadmill belt.

challenging for the patient. Velcro weights can be placed around the carpus and tarsus to increase the work a patient must carry out to move their limbs during therapy. A source for pre-made leg weights is Canine Icer ([www.canineicer.com/](http://www.canineicer.com/)), or use curtain weights to make your own weighted wraps. Weights must be used with caution to prevent patient fatigue. Additional flotation can also be added to a patient's lower limbs using either children's air-filled water wings, modeling balloons, or foam piping insulation cut into short strips. The added flotation means the patient must use more effort to bring their paw down to the treadmill belt during movement as the added buoyancy draws the limb upwards.

## Safety and Maintenance

Microbiological contamination of UWTM and pool water can result in pathological organisms causing infections in both patients and therapists. Contamination is introduced into the water largely by our canine patients, with skin and fecal bacteria causing most bacterial contaminations. Pathogens can also be introduced into the water from poorly maintained water filters and occasionally from defects in the pool design. The regional legislation regarding water quality in UWTM and pools should be consulted to ensure local health and safety legislation is being adhered

to. Microbiological assessment of UWTM and pool water is largely directed at identifying markers of fecal contamination and assessing the levels of *Pseudomonas aeruginosa*, as these bacteria have been associated with ear and skin infections in both dogs and therapists. The presence of significant colony numbers of *Ps. aeruginosa* during microbiological testing or the presence of *Escherichia coli* indicates poor water management and remedial action must be taken.

Water management is an integral part of running a safe aquatic therapy unit. Poor water quality or inappropriate use of water management chemicals can have a major impact of the health and safety of therapists and patients. Poor water quality can also lead to staff sickness, staff injuries, and staff absences. From the patient's point of view, poor water quality can also lead to sickness and injury which will lead to client dissatisfaction and have a detrimental effect on your business success. The complexity of the water management protocol will be dictated by the size of the UWTM or pool and the number of patients using the pool daily. Protocols for water management should be tailored to meet the exact needs of the unit and there must be records outlining when, and by whom, water management tasks are carried out.

All owners should be instructed to bring their pet to an area to encourage defecation and urination prior to commencing an

aquatic therapy session. Despite the best efforts of owners and therapists our canine patients will challenge the therapist from time to time by defecating during an aquatic therapy session. Fecal contamination of the pool water must be dealt with swiftly and safely. Nets and sieves can assist in removing formed fecal matter and, if carried out in a timely fashion, the UWTM or pool can likely be kept open. If loose feces are passed in the UWTM or pool, it is likely the unit must be shut down for 3–4 hours and a decontamination cycle run to maintain water safety. This highlights the importance of the owner answering pre-hydro questions regarding the recent toilet habits of their pets. In the case of patients who have fecal incontinence or who are “repeat offenders” contaminating the water, manual evacuation of the rectum and a bath before hydrotherapy is recommended.

## Protecting Staff and Clients

When working in and around pools and UWTM all staff and clients must be made aware of the risks of being in an aquatic therapy unit. Local legislation regarding health and safety around a pool or UWTM should be carefully researched, including the appropriate guidelines regarding safe working practices and rules within the unit to ensure the safety of client, patient, and therapists. There should be clear protocols to deal with emergencies such as clients becoming unwell and it is advisable that staff members are trained in first aid techniques to assist both their colleagues and clients in the event of an injury on the premises.

## Pre-Hydro Patient Assessment

### Initial Patient Assessment

Initial patient assessment should occur prior to the patient arriving for hydrotherapy. Paper or email referral forms should be sent

to both the client and the primary care veterinarian. Question the client to ascertain a patient's prior experience with water. Patient and therapist safety is paramount and dogs that have a serious fear of water or are reluctant to swim may pose a danger to themselves and to therapists. When a patient panics, the natural reaction of a therapist is to try to restrain the patient and remove them from the water. This can lead to injury and, in extreme cases, the loss of bite inhibition caused by panic may lead to the therapist receiving serious bite injuries. Low-stress handling should be used and slow introduction to the environment performed. Client questionnaires should also ask about the best way to motivate a patient (e.g., food or toys) and establish whether the patient is safe to be around other canines. The questionnaire should be accompanied with a set of therapy guidelines, health and safety guidelines, and terms and conditions. The veterinary referral form should focus on why the patient is being sent for aquatic therapy and what the supervising rehabilitation veterinarian is hoping to achieve from therapy. There should be declarations regarding coexisting diseases that may interfere with aquatic therapy and the patient's vaccination status should be confirmed on the referral form.

### Client Questionnaires

Upon arrival at the therapy unit the client should be asked a standard set of questions which are designed to assess the patient's immediate suitability for aquatic therapy. The pre-session questions should ascertain if the patient has emptied their bladder and bowels prior to the session. The therapist should also focus on identifying how the patient was in the hours after the last aquatic therapy session. This information should be used to form a basis for whether the therapy session needs to be made more challenging.

### Coexisting Disease

Understanding why a patient is presenting for aquatic therapy and how pre-existing medical

conditions affect their ability to safely engage in aquatic therapy is vital. Preexisting medical conditions such as heart, respiratory, kidney, or liver disease do not preclude a patient from engaging in aquatic therapy. Each patient should be evaluated as an individual and a tailored plan created. It is unsuitable for a patient with concurrent clinical cardiac disease to engage in strenuous swimming or treadmill activity, but they may benefit greatly from water-based massage therapy and water-based range of motion exercises. The key to creating tailored aquatic therapy plans for patients with multiple disease processes is gaining a complete understanding of the patient's limitations and understanding the risks that water-based therapy may present to that individual. The rehabilitation veterinarian will perform a thorough examination determining the patient's abilities to carry out aquatic therapy. Gaining a complete medical and orthopedic history for every patient is key; a close working relationship with referring veterinarians is also vital for creating patient-centric treatment plans.

Surgical Wounds

Although historically, patients with surgical (sutured) wounds were excluded from aquatic therapy, new advances in surgical wound management are challenging this concept. There are a variety of wound sealant products that can be applied in theatre to theoretically completely seal a surgical wound to allow early aquatic rehabilitation (Baranoski and Ayello, 2012). The performance of such products has been evaluated in human medicine and studies will continue in animals to validate the use of these products in animals receiving aquatic therapy in the early postoperative period. Patients with open wounds or infected surgical wounds should not receive aquatic therapy until the wound is fully healed.

Contraindications for Hydrotherapy

Although every effort should be made to create a treatment plan that accommodates a patient's coexisting disease, there are several

presentations that exclude a patient from aquatic therapy either temporarily or permanently. For the protection of staff members, and in the interest of water hygiene, dogs with acute or chronic gastrointestinal disease that leads to either vomiting or diarrhea should be excluded from aquatic therapy and should see the veterinarian. Patients who are obviously unwell with evidence of an elevated body temperature or signs of malaise should not receive therapy and instead should be directed to their supervising veterinarian for assessment. The chemicals used in water management have been shown to irritate damaged skin and therefore patients with acute dermatological problems (e.g., acute moist dermatitis) should not swim (Nganvongpanit and Yano, 2012). Patients with chronic dermatological conditions should only swim under the clearance of the supervising veterinarian and attention should be paid to post-therapy management (Box 20.5). Patients with signs of infectious diseases, such as kennel cough, should be excluded from the clinic until the infectious period has passed. Lastly, patients with a history of uncontrolled epileptic activity should not engage in aquatic therapy until their condition is medically controlled.

There are patients that you will encounter who are simply not suitable to engage in aquatic therapy. Although it is recommended

**Box 20.5 Contraindications to hydrotherapy/aquatic therapy**

- Unhealed surgical incisions
- Skin irritation or infection
- Emesis
- Diarrhea
- Untreated cardiac, liver, or kidney disease
- Incontinence
- Uncontrolled epilepsy
- If the dog has an external fixator, as there is a risk of infection associated with open wounds

*Source:* Adapted from Tomlinson (2013) and Connell and Monk (2010).

to try at least 2–3 sessions to acclimate and get a patient used to either pool-based or UWTM activity, some patients find the experience too stressful or use their limbs in a counterproductive fashion and are at risk of making their particular condition worse.

### Managing the Incontinent Patient

Opinions will remain divided on whether the urinary incontinent patient should receive aquatic therapy due to the risk of pool contamination and ascending urinary tract infection. The most commonly encountered urinary incontinent patient is the neurological patient suffering from intervertebral disc disease. Aquatic therapy plays a key role in the rehabilitation of these patients and withholding such therapy could prolong the patient's recovery time. Several strategies can be applied for managing the urinary incontinent patient. The first involves pre-therapy bladder emptying to ensure the patient has an empty bladder prior to starting aquatic therapy. Technicians/nurses who have received veterinary training in bladder expression can empty the bladder using manual expression. This should provide a 30- to 60-minute window where the patient will produce a negligible volume of urine and should be able to receive UWTM or pool-based rehabilitation without urine contamination. Specially designed canine incontinence pants can also be used to minimize leaking urine. Infant swimming diapers can also be used in small dogs.

Patients with urinary incontinence can also be managed with a Foley catheter *in situ*. This remains a controversial strategy, with some believing it predisposes the patient to ascending urinary tract infection. If the catheter is meticulously managed during aquatic therapy, there should be no reason why its presence should prevent a patient receiving therapy. The bladder should be actively emptied via the Foley catheter prior to therapy. The catheter should then be sealed using an appropriate bung to prevent water entering. At this point the patient can begin therapy.

On completion of therapy, the external portion of the Foley catheter and the prepuce should be flushed and cleaned with dilute chlorhexidine solution. The bung should be disposed of and the patient reconnected to a closed collection system.

Patients with fecal incontinence may be presented for aquatic therapy. It is the author's opinion that these cases should not receive aquatic therapy unless an appropriate swim diaper is placed after evacuation of the rectum. Many times, patients can have a bowel movement stimulated by using a "Q-tip" around the anal sphincter. This helps to stimulate evacuation. If a patient has persistent liquid feces or cannot be managed using a swim diaper the patient must not swim because of the risk to the health and safety of the therapist.

### Patient Conformation

Although almost any size and shape of dog or cat can be taught to swim in a pool or use an UWTM there are some considerations that should be made for individual breeds, and the challenges that their specific conformation and breed personality present to the therapist. The conformation, or size and shape of a patient, affects a patient's ability to swim. Brachycephalic breeds, such as Bulldogs and Pugs, with shortened faces, have narrowed nasal passages. This conformational change is often accompanied by an excessively long soft palate and a small trachea in comparison to overall body size. These changes restrict these patient's athletic capability as they simply cannot effectively move enough air in and out of their lungs to maintain strenuous athletic exertion. These dogs may also struggle to carry toys in their mouth during therapy and protect their airway at the same time. With these breeds, careful attention should be paid to providing enough flotation in the form of jackets and ancillary flotation devices to maintain the patient's head above the surface of the water always. The therapist must also pay careful attention for the early signs of fatigue.

## Pool Versus Underwater Treadmill

There are distinct differences in the therapeutic benefits of pool-based aquatic therapy and those of UWTM rehabilitation. The choice of pool-based therapy or UWTM therapy is tailored to the problem that is being treated. Many patients can benefit from a combination of pool-based therapy and UWTM therapy. It should be remembered that many small and medium patients can swim in the UWTM.

Pool-based therapy allows the patient to be rehabilitated without bearing any weight; this is ideal for patients who are painful when weight bearing, or are unable to weight-bear due to neurological disease or an unstable limb (Figure 20.7). Rehabilitating patients in a non-weight-bearing environment can be continued until these patients start to show improvement; they can then be transitioned to UWTM therapy to help



**Figure 20.7** Toys make excellent motivational tools in the pool. Care should be taken to avoid airway impedance.

retrain the patient to gait normally and to slowly return to weight bearing. Patients who lack flexion when gaiting on land can benefit from swimming versus UWTM activity as the limbs are in flexion during swimming (Marsolais *et al.*, 2003).

The water height in a UWTM can be adjusted to tailor the amount of weight a patient is bearing during therapy (Levine *et al.*, 2002, 2010). As a patient improves during therapy the water level can be lowered to allow more weight bearing, and then upon completion of the therapy program the water can once again be deepened to allow the patient's athletic capabilities to be challenged (Box 20.6).

The UWTM provides some distinct advantages over pool-based therapy in certain scenarios and vice versa. Most UWTM have clear plastic sides which allow the patient's gait (and joint extension and flexion) to be assessed during motion. This can be very difficult to assess when using pool-based therapy. UWTM provide a more controlled environment in which to introduce animals that have never encountered water to paddling and even swimming. Patients bear weight during training on an UWTM, which

### Box 20.6 Assessment of anxiety and fatigue when swimming

#### Anxiety

- Anxious facial expression (lips pulled back)
- Rapid breathing/pulse
- Thrashing forelimbs
- Inability to rest
- Attempting to exit the pool

#### Fatigue

- Tired body posture
- Deep or irregular breathing
- Slowed swim pace
- Reluctance to swim
- Change in tongue color/shape

Source: Chiquoine *et al.* (2013). Reproduced with permission of John Wiley & Sons.

may benefit patients that are recovering from fracture repair, tendinopathies, and other conditions in which weight bearing can stimulate healing. The controlled weight bearing offered by a UWTM can be advantageous during rehabilitation of fracture patients or those recovering from joint instabilities. Pool-based therapy is more suited for strengthening a patient's core and trunk muscles as the rotational forces encountered during swimming help build these muscle groups. A pool allows the patient's whole body to be treated and enables massage and manual therapy to take place in the water environment; this is very difficult to do in a UWTM because of space limitations.

### Pool-Based Exercises

#### Static Therapy

Static therapy ideally takes place on a platform within the pool, allowing as much of the patient's body to be submerged with the patient's paws on a solid surface. Static therapy is used to help treat specific areas of muscle pain, improve patient comfort, and increase the range of motion of a joint. In addition, static pool-based therapy helps re-educate patients in maintaining a normal standing posture. It should be noted that although patients are often referred to treat a specific joint or problem, it is imperative to take a holistic approach to aquatic therapy (the meaning of holistic being to look at the patient's whole body). For example, a patient recovering from cruciate ligament surgery may have muscle fatigue and pain in the contralateral non-operated limb due to transitioning weight in adapting to operative limb pain and weakness. Water-based massage therapy can help to reduce muscle pain, reduce postoperative edema, improve flexibility, and can greatly improve a patient's comfort and wellbeing.

The pre-therapy examination should focus on identifying areas of pain, discomfort, and tightness in major muscle groups. Therapy should then focus on massage of these muscle groups by a qualified individual along

with gentle stretches aimed at improving flexibility (see Chapter 5). Joint mobilization aims to improve joint motion and reduce joint pain, thus improving joint function. Carrying out joint mobilizations underwater adds the warming and hydrostatic benefits of the aquatic therapy to this modality. It should be noted that joint mobilizations should only be carried out by suitably qualified rehabilitation veterinarians or physical therapists. Postural exercises such as three-legged standing can be used in the pool to encourage a patient to actively load a specific limb.

#### Active Therapy

Pool-based active aquatic therapy aims to improve muscle strength, improve aerobic fitness, increase flexibility, and improve range of motion. It tends to involve either swimming lengths or laps of the pool depending on the size of the pool. In an endless pool the patient will swim against the water flow, which provides resistance. In the early stages of therapy, the patient must be supported and guided by a technician/nurse. Active therapy does not mean simply allowing a patient to swim uncontrolled. An active role is taken by the handler during the session and ensures the patient is using their limbs as normally as possible during therapy. Patients who have limb paresis and paralysis have their affected limbs cycled through a swimming motion (active assisted range of motion) during therapy to stimulate neural pathways and encourage volitional limb movement. In large patients two handlers may be required—one to guide and support the patient and a second to carry out the therapeutic exercises. Toys make excellent motivational aids for patients engaging in pool-based aquatic therapy and are often a necessity to ensure patients engage with therapy and get the most out of a session.

During assisted swimming the technician/nurse can challenge the patient by gently rolling the patient from side to side while swimming. This rolling movement, described as a perturbation, stimulates the patient's righting reflex. Stimulating this postural reaction evokes limb

extension mediated through neural feedback pathways with stepping responses in the patient, which, when translated onto land, helps patients with balance, bending, and turning. These perturbations also help build and strengthen a patient's core stabilizing muscles. A point may be reached with some patients where little support or guidance is required during swimming due to the patient having a high level of proficiency. A technician/nurse should remain in the pool with these patients, however little input is required.

The patient that becomes adept at swimming can be transitioned to a lake and swimming with companions (Figures 20.8 and 20.9). This technique can be used for maintaining cardiovascular fitness.

### Treadmill-Based Exercises

#### Speed

There is a lack of research to determine exactly what belt speed is most appropriate for patients with specific conditions and for

specific breeds. However, there are a wealth of experienced aquatic therapists that have established the most useful treadmill belt speeds for a variety of breeds and conditions. Leg length and stride length must be taken into account when deciding upon a belt speed. It should be noted that smaller dogs often walk relatively faster than larger dogs (Voss *et al.*, 2010). It was stated earlier that treadmill belt speed is not calibrated to miles per hour (or km/h) by most UWTM manufacturers, therefore advice regarding speed is not given in this chapter. It is best to say the speed should be adjusted to the most comfortable-looking gait in each patient, whether that gait is walk, trot, or, in some cases, faster. Pacing should be avoided unless the dog has always paced naturally. Doubling belt speed from the walk should in most patients initiate a trotting gait. If a patient fights the transition into a trot it may start to use a pacing



**Figure 20.8** Swimming in open water – in this case, a lake.



**Figure 20.9** If multiple dogs are entering the pool/lake at the same time during a fun swim session it is important that all the dogs are confident in the water, and are wearing appropriate jackets or other flotation aids.

gait. Patients can be moved out of a pace into a trot by encouraging them forward on the belt, using a leash or toy.

Patients with neurologic issues may have a dissociative gait where the forelimbs move at faster tempo with shorter strides than the rear limbs (Gordon-Evans *et al.*, 2009). A platform can be added over the front of the treadmill belt so that the patient can stand with the forelimbs and gait with the rear limbs. This has been successful in practice for gait retraining these types of patients (Tomlinson JE, personal communication).

### Water Depth

The depth/height of the water has a great impact on the amount of work being carried out by the patient during UWTM aquatic therapy. Studies have examined how water depth affects the amount of weight a patient is supporting during therapy. Filling the UWTM to the level of a patient's tarsus reduces the weight bearing of the patient by 9% (Levine *et al.*, 2010). Weight bearing is reduced by 15% if the water is filled to the level of a patient's stifle. If the treadmill is filled to the level of a patient's great trochanter (hip joint) we see the greatest reduction in weight bearing, with a maximal reduction in total weight bearing in the region of 62% (Levine *et al.*, 2010).

Filling the UWTM to different levels on a patient's body does not only affect the total weight bearing, it also influences the ratio of weight being supported by the forelimbs in comparison to the hindlimbs. Unaided by the buoyancy of water, dogs support approximately 60% of their weight through their forelimbs. This is largely unchanged as water is brought up to their hock and stifle joints (Levine *et al.*, 2014), but as water is filled up to a patient's great trochanter we see the forelimbs supporting up to 71% of the patient's weight (Levine *et al.*, 2010). Remember that overall weight bearing is reduced by 62% when the water is this high, but this weight redistribution may still be of significance in patients who have forelimb disease as UWTM therapy with deep water levels may in some patients exacerbate forelimb weight load.

Patients who have an adaptive weight shift to the forelimbs due to hindlimb insufficiency may be better served by being exercised in the UWTM at a maximum water height of the stifle to avoid exacerbating the forward postural shift.

### Water Temperature

Pool and UWTM water temperature is an important variable to control during aquatic therapy. Studies have investigated the ideal water temperature for patients undergoing therapy. Water temperatures between 30 and 34°C were examined and the effects of different temperatures on heart rate, respiratory rates, rectal temperature, and patient exertion were measured. Research showed that all patients undergoing therapy showed increased vital parameters during therapy, but there were no significant differences in parameters between different water temperatures. This research demonstrates that water temperatures between 30 and 34°C (86–93.2°F) are considered safe for canines (Nganvongpanit *et al.*, 2014a).

Maintaining the water temperature of a pool or UWTM represents one of the largest expenses for an aquatic therapy business and water temperature should be thermostatically controlled to control energy expenditure.

### Inclination

Exercising on an incline on land changes joint kinematics in dogs (Bockstahler *et al.*, 2012). Increasing the incline of the UWTM belt during therapy likely increases the work for the patient but the effects on joint motion are unknown. Most neurological and orthopedic patients will not require the addition of an incline to the UWTM belt during therapy, but those patients who have recovered fully from their injury and canine athletes may benefit from the additional of an incline to the belt during therapy to challenge their fitness during therapy.

### Session Duration

It is imperative to build the duration of activity in a patient's aquatic therapy slowly.

A patient's first few UWTM sessions may not progress beyond 2–3 intervals of walking for 45–60 seconds. Patients should be given 90–120 seconds to recover by standing still in the water between training intervals. This recovery time frame may need to be shortened in animals with a high level of fitness or lengthened in very unfit animals. As a patient becomes more confident on the UWTM the interval length should be increased session by session. At the end of a session a patient should be evaluated to determine how difficult they found a session. If they are excessively exhausted by the session it should be made less intense at the next appointment. If the patient obviously coped with the session well it should be made tougher at the next appointment.

## Developing Patient Programs

The patient's first session in the pool or on the UWTM is often more of an acclimatization session rather than a true therapeutic session. It is important to get the patient as comfortable as possible and get them used to the sensation of water moving up their body as well as to the movement of the treadmill belt prior to engaging in useful therapy sessions. Engaging in static therapy in the pool or UWTM during the first session is a very useful acclimatization tool (Figure 20.10). A whirlpool bath can be used for the neurological patient at the beginning of rehabilitation therapy.

A patient's treatment program must have specific goals. The goals of a program dictate the ideal number of sessions a patient should engage in on a weekly basis. Patients early in their recovery from a neurological or orthopedic injury benefit from more frequent sessions. Both UWTM and pool-based therapy revolve around repetition of training intervals. In the pool an interval may be a single length or lap of the pool. As a patient progresses through an aquatic therapy program,



**Figure 20.10** Whirlpool bath for a neurologically impaired dog.

the number of repetitions will be slowly increased during the session, while at the same time the rest period between intervals is slowly decreased. In the UWTM, the number of intervals for a given time is increased, as is the length of the training interval as the patient progresses in an aquatic rehabilitation program.

There are some parameters that can be examined to determine if a program is meeting the goals for a patient. The recovery of a patient in the hours after a pool or UWTM session is one of the best guides of whether a program is suitable or requires adjustment. If a patient recovers rapidly and does not appear tired after a rehabilitation session, it is possible that the session was not demanding enough and at the next session the duration or speed should be increased. When a patient appears tired for 4–6 hours after a session one must consider that the session was too demanding for the patient and the next therapy session should be adjusted so that the session is less demanding on the

patient. This advice is all subjective and you should be guided by your supervising veterinarian as fatigue needs to be carefully avoided in some patients.

Once rehabilitation has been completed the patient should ideally be transitioned to a maintenance program that focuses on maintaining the improvements that have been achieved. This maintenance program may include hydrotherapy. Maintenance therapy should consistently bring a patient to the edge of their ability to maintain cardiovascular fitness, preserve muscle mass, and, most importantly, retain the improvements gained during an aquatic therapy program.

Canine athletes are common visitors to aquatic therapy centers, not only to treat specific orthopedic problems but also to maintain and improve cardiovascular fitness during the off-season to maintain fitness and improve athletic prowess. Once a patient can easily achieve a level of continuous exercise in a pool, it can be difficult to apply further challenges to patients in the pool apart from increasing the length of the session. The UWTM may be a more versatile tool for canine athletes aiming to maintain fitness, as it allows an array of parameters to be adjusted to challenge the patient during their session. Resistance jets (may also be available in some pools), incline, water height, and speed can be adjusted. The aim for canine athletes who are engaging in UWTM should be to reach a point where they can maintain a single interval on the UWTM of between 20 and 30 minutes at a brisk walk without a break. Once a patient can maintain a brisk walk for 20–30 minutes the session should be interspersed with a small number of brief 20- to 30-second intervals at the fastest pace the canine can maintain.

### Business Diversification

There are significant costs associated with running a UWTM or swimming pool. To maximize income, any rehabilitation center offering these therapeutic options should examine how they can make these assets pay

for themselves. Adapting opening hours to suit client needs is a basic strategy that can help increase bookings; evening and weekend openings offer appointments outside of the usual daily working hours and are very popular. In addition to flexible opening hours, diversifying away from pure aquatic rehabilitation and exercise therapy can assist in increasing UWTM/pool revenue and ensure maximal use of the facilities.

### Obesity Clinics

While aquatic therapy is unlikely to be successful in achieving weight loss in the obese canine/feline if used alone, if combined with dietary restriction and lifestyle modifications it can be a very effective means of achieving weight loss. The veterinary technician or nurse should remain in the UWTM or pool with the patient (Figure 20.11). Accelerated weight loss has been reported in patients that have hydrotherapy incorporated into a weight loss program. Weight loss of up to 1.5% body weight per week was reported in a study that saw patients exercising on an UWTM for up to 30 minutes once a week



**Figure 20.11** Technician Jenn Panko in the underwater treadmill with a feline patient. Even when little support or guidance is required due to the patient having a high level of proficiency, a technician/nurse should remain in the underwater treadmill/pool with these patients.

(Chauvet *et al.*, 2011). This is compared to reported weekly weight losses of 0.75% per week with dietary modification alone.

If you are going to promote UWTM or pool swimming as a weight loss tool, it is important that you involve the supervising veterinarian in the development of the exercise program as well as details of dietary management. Client education, dietary management, and additional home-based exercises are key to successful weight loss and one cannot rely on aquatic therapy alone. Owner compliance will be better if results are seen early and these results can only be achieved if a holistic approach to weight management is taken. Once a target weight has been achieved, owners can be encouraged to continue aquatic therapy on a maintenance basis to help the patient maintain their target weight.

#### **Developing the “Fun” Swim**

The non-therapeutic aquatic session or “fun swim” can prove an important revenue stream for rehabilitation centers with a pool. Fun swims need to be supervised to ensure dog and client safety, but they are less labor intensive than therapeutic sessions. If multiple dogs are entering the pool at the same time during a fun swim session it is important that all the dogs are confident in the water, they are wearing appropriate jackets or other flotation aids, and that there are sufficient staff members to evacuate the animals from the pool in a timely fashion. Legal disclaimers need to be incorporated in the United States.

#### **Puppy Water Introduction Classes**

As previously discussed, there are a subset of dogs that have not experienced swimming or being partially submerged in water. This can lead to fear and aversion to water in dogs. Puppy water introduction classes are aimed at exposing young dogs to water in a controlled and safe fashion in either a UWTM or a pool. Buoyancy aids should be worn in the early sessions. Some dogs can be safely introduced to water over a single session, but

some may require multiple sessions to become confident in water. One should first introduce a puppy to water of a depth level to the patient's carpus and use positive reinforcement, either verbal or treat based, to put the dog at ease. The water should be gradually deepened over the session while ensuring the dog feels comfortable and reassured. This is where working “pods” in a pool come into their own, as you have a static platform on which to work with the dog. Over the water introduction session in a pool with a platform, the dog should be encouraged to lift its feet off the platform and begin to swim. If the dog becomes distressed, return to a solid surface and work on building the dog's confidence prior to attempting swimming again.

#### **Water Safety Courses**

Older dogs can be safely introduced to water in a similar fashion to puppies. Water safety classes are something that can also be utilized in working dogs, such as army and police dogs, who currently rarely receive water safety lessons during their training. Police and army dogs, as well as other working dogs such as gun and trial dogs, may unexpectedly enter water of varying depths in their working life and should be trained to cope with this. Handler involvement is paramount to success in effectively training working dogs and if insurance permits, the handler should be directly involved, ideally being in the pool with their working dog. One particular area of training that can assist working dogs is training dogs to exit water unassisted, which could prove a life-saving skill if a working dog enters water without supervision.

### **Promoting Your Business**

Successful promotion of an aquatic rehabilitation business can be the key to accelerating a rehabilitation clinic's growth and the onus for this frequently falls on the veterinary nurse/technician rather than the supervising veterinarian. All investments in advertising

should focus on reaching clients and referring veterinary surgeons within a sensible geographical radius. Investing in a quality waterproof camera will help you generate engaging and interesting images and videos to use in promotion of the business. Underwater videos can also be used to document a patient's session-by-session progress.

Building relationships with referring veterinarians is key to maintaining a steady flow of rehabilitation cases and a lot can be said for visiting possible referring clinics to discuss what you can offer. Face-to-face communication can build business relations that can support the growth of a rehabilitation center in its early years and builds security by providing a steady stream of referrals. Having a physical presence at local community events and sponsoring local community teams and events all helps to build community bridges that builds the clinic's profile. Your veterinarian should join you.

Social media provides a cost-effective platform through which you can engage with clients, potential clients, and referring veterinary surgeons. A Facebook business page can provide a fledgling aquatic therapy unit with an excellent platform on which to advertise the business prior to making the investment in a webpage and domain. Publishing regular, interesting, and succinct articles about the conditions that aquatic therapy can assist

with is an excellent starting point to gain confidence in writing and publishing social media content. Twitter can be used in a similar way to Facebook to generate client interest and promote aquatic therapy. Encouraging current clients to engage with your social media platforms is paramount if a venture into social media is going to work. Publishing daily updates about the patients you are treating is an excellent way of growing interest in an aquatic therapy business. When venturing into social media it is vital that client confidentiality is protected always and permission needs to be gained from all clients prior to publishing pictures of their pet, or any images of the actual client.

## Conclusion

Aquatic therapy remains a cornerstone of rehabilitation for many orthopedic and neurological conditions. When the approach taken to therapy is regimented, individualized, and responsive to a patient's progress and when it considers the holistic needs of the patient, the impact on a patient's life can be hugely positive. This sphere of therapy will continue to mature as research helps us to define specific benefits that aquatic therapy has for a spectrum of specific conditions.

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## 21

## Troubleshooting as a Team

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## Introduction

The veterinary rehabilitation team works towards a common goal: improving quality of life, relieving pain, and maximizing function for a patient, even in the light of physical impairments. The veterinary rehabilitation team can consist of a rehabilitation veterinarian, a rehabilitation veterinary technician, and a physical therapist trained in animal rehabilitation. Adjunct team members can include veterinarians trained in acupuncture and manipulative therapies, chiropractors with further training in therapy on animals, massage therapists, veterinary assistants, and physical therapist assistants. Collaboration with primary care veterinarians, veterinary specialists (neurologists, surgeons, oncologists,

etc.), other veterinary clinic staff, and the owner/handler/caregiver is integral to the success of rehabilitation. This collaboration is especially vital when troubleshooting patient problems or lack of progress.

### The Rehabilitation Veterinarian

The rehabilitation-trained veterinarian will examine the patient and make a diagnosis including assessment of pain level, injured tissues, concurrent disease and functional limitations including strength, range of motion, balance and proprioception, and gait impairment. If further diagnostics are needed, they will be performed. The rehabilitation veterinarian will develop a treatment plan including at-home and in-clinic therapies,

medications, and recommended nursing care. The veterinarian may perform manual therapies. Other professionals, if present, will contribute to this plan. The rehabilitation veterinarian may be a specialist in rehabilitation and sports medicine (DACVSMR) or certified in rehabilitation.

### **The Rehabilitation Veterinary Technician**

The rehabilitation technician will assist the rehabilitation veterinarian, physical therapist, and any other adjunct professional with the examination. The technician will assist in handling the patient, taking and recording measurements and observations, such as vital signs and pain scoring. The rehabilitation technician will take part in educating the client and explaining the home care and in-clinic therapy plans. The rehabilitation technician will also take part with in-clinic therapies under the supervision of the rehabilitation veterinarian. The rehabilitation technician should have received specific training in (if not certification in) veterinary rehabilitation.

### **The Physical Therapist**

The physical therapist, trained in animal rehabilitation, will evaluate the patient with the rehabilitation veterinarian and document impairments in range of motion (including accessory joint motions), flexibility, strength, functional mobility, gait, and neurological function. Together with the rehabilitation veterinarian, the physical therapist will develop and institute the treatment plan and will perform treatments including manual therapy, facilitation techniques, therapeutic exercises, and physical modalities. Physical therapists who work in animal rehabilitation need to be certified in such, or hold a master's degree in animal rehabilitation depending on state legislation. A physical therapist trained in animal rehabilitation may work with and supervise a physical therapist assistant. This is an individual who has an associate's

or bachelor's degree and can provide some therapies under supervision of a physical therapist. As with all team members, the physical therapist assistant should have received training in animal anatomy, biomechanics, handling, and rehabilitation.

### **Other Team Members**

Other team members can include veterinarians who are trained in acupuncture or spinal manipulative therapy (veterinary term for chiropractic) and other veterinarians including specialists (surgeon, oncologist, neurologist, primary care veterinarian). Chiropractors who have received additional training and certification in animal chiropractic along with massage therapists may be involved in the care of the rehabilitation patient.

### **Case Management**

Each case will be very different; two dogs with the same problem that have undergone the same surgical procedure at the same facility will have differing levels of function, pain, and recovery rates, even when everything goes per plan. This rather obvious statement highlights the need to avoid applying a set protocol to groups of patients. The foundation of the therapeutic plan is the patient evaluation. Patient evaluation should continue throughout the rehabilitation process; evaluation of patient function, pain score, and examination for concurrent disease should occur at every rehabilitation visit. The veterinary technician should immediately report to the supervising rehabilitation veterinarian any changes that have occurred from one visit to another so that appropriate measures may be taken. The therapeutic plan is altered accordingly and might include changes in medications, progression of home exercises, and relaxation of restrictions when a patient is progressing (or healing has been confirmed). Daily rounds (Figure 21.1) in which patients are discussed (both inpatients and outpatients), and the



**Figure 21.1** Patient rounds in the clinic of one of the authors (Tomlinson) are 1 hour a day. Round sheets hold written descriptions of daily therapy plan, and digital records are accessed for last therapeutic notes as each case is discussed between doctors and technicians.

therapy plan adjusted accordingly, are strongly recommended. Alerting the supervising veterinarian regarding patient concerns in the clinic may take a few moments as they may be overseeing several patients at one time. Owners should be counseled of the need to wait for veterinary intervention before progressing with therapy for that day. Troubleshooting should always involve the supervising veterinarian as the veterinarian technician is not the case manager and should not be determining the therapy plan.

### Communicating with Specialists and the Primary Care Veterinarian

It is essential that communication with the primary care veterinarian be on a frequent basis, and in the case that there is a sudden change in patient status, the communication should be immediate. The rehabilitation veterinary technician may be facilitating this communication via phone, fax, or email.

Whenever a client self-refers their own pet, the rehabilitation facility staff must inform the client that communication and collaboration with the primary care veterinarian is required.

If the patient develops a new issue that requires further diagnostics or treatment, it is the responsibility of the rehabilitation veterinarian to refer the patient back to the primary care veterinarian or on to a different specialist. In the latter case, the primary care veterinarian needs to be involved in that referral. The American Animal Hospital Association Referral Guidelines ([www.aaha.org](http://www.aaha.org)) state that, "A referral should be considered when there is a need for additional expertise/advanced training, additional services to provide further diagnostic testing or an unresolved or worsening medical condition." All veterinarians involved in the care of the patient need to be apprised of the change in condition and of any subsequent referral. Communications between veterinarians and veterinary clinics involved in patient care should include updated medication lists and care notes from the rehabilitation facility.

The rehabilitation veterinary technician is a patient advocate and should note the importance in taking a role in recognizing changes in patient status. The ultimate advice is, “when in doubt, ask your supervising veterinarian.”

## Communicating with Adjunct Team Members

When many team members are involved in patient care, many points of view about such care can come into play. The team leader and case manager is the rehabilitation veterinarian, who will consider team member input on the case with respect and professional courtesy. Collaboration is a simple process when all team members are at the same facility and have direct access to the patient. Such collaboration leads to excellent patient care. Problems can arise when members of the care team are seeing the patient at different times or in different facilities; patient status can differ day to day and recommendations may be made that disagree with those made by other team members. Team members should avoid making recommendations to the owner that contradict previous recommendations made by another team member; the case should first be discussed with the team leader. This includes recommendations about nutrition, supplements, medications, home care, and rehabilitation therapy. Team members that are not qualified to make a diagnosis or recommend a treatment should avoid doing so. For example, even though a pet store employee can make recommendations on diet and supplements without legal consequences to themselves, but which may have consequences for the animal, rehabilitation team members who are not veterinarians should not do so. Supplements and food are not tightly regulated and the wrong recommendations can do harm. The team should strive to avoid the effect of “too many cooks spoiling the broth” and refrain from making unilateral decisions.

## Examples of Case Management

### Case 1: Surgical Implant Failure

Daisy, a 1-year-old Golden Retriever, was referred to Moving Paws Rehabilitation Center by her orthopedic surgeon following failure of left triple pelvic osteotomy (TPO) which had been addressed with implant retrieval and a left femoral head and neck ostectomy. Daisy had been hit by a car prior to TPO surgery and had a history of hip dysplasia. The initial TPO had been performed 8 weeks prior to initial consultation with the rehabilitation team. Due to unknown causes the TPO failed and implants were removed 4 weeks following the initial surgery.

At the rehabilitation consult, Daisy presented with a grade 5/5 (non-weight-bearing) lameness of her left hindlimb. The rehabilitation team noted severe atrophy of the left quadriceps, hamstrings, and gluteals. She had severe contracture of her left semitendinosus and moderate iliopsoas and sartorius spasms while standing (Box 21.1, red flag box 1). The soft tissue contracture severely restricted her stifle range of motion in extension. No neurologic deficits were noted. The rehabilitation veterinarian recommended that Daisy come for therapy twice a week for 4 weeks with the physical therapist and certified rehabilitation technician, after which a full repeat assessment would be made. The owners were advised to restrict all activity at home until beginning therapy. A non-steroidal anti-inflammatory,

#### Box 21.1 Red flag boxes for Case 1

- 1) Contracture can be permanent and must be addressed effectively in a timely manner
- 2) Sedation requires the presence of a veterinarian and whole veterinary team. Without this ability, assessment would be impaired and treatment could have been delayed
- 3) Quick action needed and even then the issue may not be resolved

along with tramadol and methocarbamol, were prescribed for pain relief and to reduce muscle spasm.

The following week, Daisy began therapy. Upon examination, the physical therapist agreed with the rehabilitation veterinarian's findings and instituted a therapy plan that included low-level laser therapy (LLLT) to her coxofemoral joint, therapeutic ultrasound to her hamstrings, passive range of motion of each joint, and cross friction massage to her hamstrings. The home exercise plan included passive range of motion of the left rear limb following application of heat therapy to her hamstrings for 10 minutes twice a day. Daisy's restrictions included no off-leash activity, 5-minute leash walks 3–4 times a day, and no running, jumping, or rough play.

Following the second week of therapy, the physical therapist spoke with the rehabilitation veterinarian about sedating Daisy to assess her hamstring contracture; this was to remove any conscious protecting of the area and to provide maximal pain relief during assessment of muscle limitations (Box 21.1, red flag box 2). Upon sedation, it was determined that hip range of motion was limited to 90 degrees of flexion, 110 degrees of extension, and stifle range of motion to 90 degrees of extension due to the severity of the hamstring contracture and that surgical intervention may be required. Daisy was then referred back to her surgeon for evaluation. At 8 weeks following the TPO implant retrieval, a semitendinosus myotomy was performed. The surgeon contacted the rehabilitation veterinarian and explained that, although the surgery has increased Daisy's stifle extension, he was concerned that the muscle would quickly become fibrotic and adhere down, so he recommended she return to rehabilitation.

After Daisy's suture removal, she met with the physical therapist, and goniometric measurements were taken. Left hip flexion measured 60 degrees, extension 110 degrees, stifle flexion 36 degrees, extension 104 degrees, and tarsal flexion 88 degrees. Her initial in-hospital and at-home therapy continued as originally directed. During the

following week, the rehabilitation technician noted that Daisy's left hind paw began to knuckle onto the dorsum of her foot during her walks and informed the physical therapist. Upon examination, the physical therapist found that Daisy's digital flexors were beginning to contract (Box 21.1, red flag box 3). She called the owner and recommended that Daisy be fitted with an assistive device to help to hold her toes in an extended position for weight bearing. Upon examination, the rehabilitation veterinarian determined that as the flexion of the toes could still be manually corrected and she agreed with the decision to prescribe the device. The owner was advised to continue therapy as directed until the assistive device arrived. During this time, the rehabilitation team could advance Daisy's in-hospital exercises with weight shifting that included picking up her opposite front paw, while holding the left rear paw in a normal position to promote weight bearing.

The device was delivered and applied to Daisy's paw (Figure 21.2). The therapist



**Figure 21.2** Assistive device to prevent flexion of the rear toes in stance (Therapaw).



**Figure 21.3** An example of an orthosis which would hold the foot in adequate extension for weight bearing and, in addition, allow for some stifle and tarsal motion. *Source:* Photo courtesy of Orthopets.

found that although it helped keep her digits in a normal position, Daisy was having trouble advancing her leg. The rehabilitation technician suggested that an elastic “resistance” band be connected from the bottom of the brace up to Daisy’s harness ring over her back, to help with her limb advancement using elastic recoil. The physical therapist and technician applied the band, which aided Daisy’s limb advancement. The owner was advised to increase leash walks to 10 minutes twice a day using the band. Daisy continued therapy for the next 2 weeks.

Two weeks into the new therapy, Daisy had a recheck examination with the rehabilitation veterinarian. Her examination findings included an improvement in range of motion of hip extension, stifle extension, and tarsal flexion. However, the digital flexor tendons were contracting further. Daisy was referred back to the surgeon to discuss a possible release of the left rear digital flexor tendons. The surgeon spoke with the rehabilitation veterinarian and recommended that her left flexor tendons be released due to the contracture and possible neurologic insufficiency. The surgery was performed the following day.

The digital flexor release was successful and Daisy was no longer knuckling, but would need a long-term orthosis due to her limb abnormalities. The physical therapist

consulted with a certified orthotist (CO or CPO) and the rehabilitation veterinarian and technician took several videos, pictures, and casted Daisy’s left rear limb for an orthosis which would hold the foot in adequate extension for weight bearing, and in addition, allow for some stifle and tarsal motion (Figure 21.3). Daisy adapted to her device successfully and, although her stifle and hip extension did not completely improve to the level of her right rear limb, she had enough range of motion to be highly functional. Daisy visits the rehabilitation veterinarian every 6 months for orthosis and a general mobility check.

### Case 2: Post-Surgical Rehabilitation

Chipper, a 6-year-old, intact Pit-Bull mix, was brought into Animal Friends Animal Hospital for his annual examination and the owner mentioned an acute onset of right rear limb lameness. The owner expressed no previous issues of lameness other than a mass removal from the same leg a few years prior. The mass was not further identified via pathology and the owner felt that there was no relation between the mass removal and current lameness. Upon examination and radiographs by the veterinarian, Chipper was diagnosed with cranial cruciate disease of the

**Box 21.2 Red flag boxes for Case 2**

- 1) Menisci should be examined at surgery and any tears addressed, it may have been examined but was not noted in the records received by the rehabilitation technician. Postoperatively, meniscal problems can cause ongoing, chronic lameness and should be on the list of rule-outs
- 2) Who is supervising the case? Is the technician prescribing therapy and therefore acting out of the bounds of the Veterinary Practice Act? Does the veterinarian have rehabilitation experience?
- 3) Laser therapy can reduce symptoms of pain but we need to know the source of the pain. Why is Chipper not progressing well? Prescribing a new therapy needs veterinarian input
- 4) Owner concerns about symptoms should always be referred to the supervising veterinarian

right stifle with subsequent instability. A standard tibial plateau leveling osteotomy (TPLO) was performed. Medical notes did not state the meniscal condition (nor any treatment of the meniscus) (Box 21.2, red flag box 1). The owner was advised to restrict Chipper's activity to leash walks to the bathroom and to ice his stifle three times daily until his suture removal.

At the time of suture removal, the owner was concerned about continued lameness. The veterinarian noted that he had a grade 4/5 right hindlimb lameness and atrophy of right gluteal, quadriceps, and hamstring muscles. The veterinarian advised the owner to meet with one of his certified technicians who was also a certified canine rehabilitation assistant. During that appointment, the rehabilitation technician met with the owners and examined Chipper. She noted that there was still inflammation of the distal incision with mild restriction in right stifle flexion.

After speaking with the veterinarian, the technician recommended that Chipper come in for twice weekly in-clinic therapy. Therapy

included LLLT of the right stifle joint and quadriceps and hamstring muscles; neuromuscular electrical stimulation (NMES) of right quadriceps and hamstrings (to promote co-contraction); range of motion exercises of the right rear limb (flexion and extension of hip, stifle, tarsal, and toe joints); and introduction weight shifting and rhythmic stabilization exercises (Box 21.2, red flag box 2).

The following week, the rehabilitation technician noted that Chipper's lameness had improved and advised the owner to begin 5-minute slow leash walks 3–4 times daily along with passive range of motion of the right stifle in flexion and extension followed by icing twice daily. Chipper had nearly completed the prescribed postoperative course of anti-inflammatories and cefpodoxime proxetil. Cefpodoxime proxetil is an oral cephalosporin antibiotic used to treat bacterial infections of the skin, such as wounds and abscesses as well as bladder and respiratory infections in dogs and skin and soft tissue infections in cats.

After 3 weeks of therapy, the owner met with the rehabilitation technician and expressed concern that Chipper was not progressing well at home, although they had noticed him feeling better the day following laser therapy. The rehabilitation technician recommended that the owner bring Chipper in for laser therapy an additional day during the week (Box 21.2, red flag box 3). As Chipper continued to come in for therapy, the rehabilitation technician continued NMES along with weight-bearing exercises, but noted that Chipper's progress was slower than expected.

Another 2 weeks of therapy had passed and the owner was very concerned (Box 21.2, red flag box 4) that Chipper was still toe touching and "not really wanting to use his leg." The rehabilitation technician recommended that the owner meet with the veterinarian. Upon examination, the veterinarian noted severe atrophy of right gluteal, quadriceps, and hamstring muscles with a grade 3/5 lameness. Since Chipper was 8 weeks post-op, the veterinarian took radiographs to ensure that the surgical plate was stable.

The radiographs showed that the plate was in the proper position, but unusual, reactive bone remodeling was seen distal to the plate. The veterinarian recommended and performed a bone biopsy. Chipper was referred back to continue therapy until the results came in. The next morning, however, the owner called and reported that Chipper's leg had become swollen and they had to take him to the emergency room. The emergency room doctor repeated radiographs, and saw a pathologic fracture secondary to cancerous remodeling. Chipper was referred to an oncologist. His diagnosis was osteosarcoma and his prognosis poor as it had already metastasized to his lungs.

What are some of the things that could have been done differently with this case? Be sure to note that osteosarcoma can cause lameness that starts mild and progresses, and is also associated with previous surgery sites (Holmberg *et al.*, 2004; Selmic *et al.*, 2014).

### Case 3: Activity Restrictions

Luke, a 4-year-old, neutered, Belgian Malinois was referred to Fit Furry Friends Animal Rehabilitation Center, 6 weeks after surgery to repair an Achilles tendon rupture. Initially, Luke had presented to the orthopedic surgeon following 4 weeks of non-weight-bearing lameness after a laceration to his left

tarsus. The laceration had been repaired by his primary care veterinarian, but a week after surgery he began walking with an almost fully plantigrade stance (Figure 21.4). Surgery was performed and the identified tendon tear was apposed with sutures. The surgeon noted that there was an increased amount of tension on the repair since the tendon had retracted proximally. A calcaneotibial screw was placed to lock the tarsus in hyperextension during healing. A full bivalved cast was also placed over a bandage to restrict movement of the stifle and distal limb. Luke returned for weekly bandage changes until 4 weeks after surgery, when removal of the calcaneotibial screw was performed and his cast was changed to a soft padded bandage with a lateral splint. Bandage changes continued weekly for 2 weeks, at which time he was referred to the rehabilitation team for orthosis prescription and rehabilitation consultation (6 weeks after surgery).

During Luke's initial consultation with the physical therapist, he was non-weight bearing on his left hindlimb. He had moderate atrophy of his left gluteals, quadriceps, and hamstrings with severe restriction of left tarsal flexion range of motion. After taking measurements and a fiberglass impression for an articulating tarsal and foot orthosis (Figure 21.5), the physical therapist consulted



**Figure 21.4** A dog with an almost plantigrade stance of the left rear due to Achilles tear.



**Figure 21.5** An articulating tarsal and foot orthosis (Orthopets) placed on the patient in Figure 21.4.

with the surgeon regarding the implementation of other modalities that would help facilitate Luke's healing. The surgeon was not a "fan of rehab" and elected to keep Luke in a soft padded cast and declined any further therapies at present (Box 21.3, red flag box 1).

Two weeks later, Luke returned for his orthosis fitting and adaptation exercises. Upon consultation with the surgeon again,

the physical therapist was advised not to begin additional therapies. The rehabilitation technician reviewed orthosis care instructions and recommendations with the client, who was advised to return to the surgeon for a recheck (Box 21.3, red flag box 2). The following week, the surgeon contacted the physical therapist and reported that a skin sore had formed, so the surgical staff replaced the orthosis with a soft padded lateral splint. The owner was frustrated with this "step back" in healing and wanted more assistance with orthosis use once the wound had healed. The surgeon then contacted the physical therapist and advised that it would be acceptable to initiate rehabilitation modalities and take over care of the orthosis after a week in the bandage splint to allow sores to heal.

At the next rehabilitation visit, the bandage splint was removed, the orthosis was reapplied (Box 21.3, red flag box 3) and LLLT of the left Achilles tendon, quadriceps, and hamstrings was initiated. NMES was applied to the quadriceps and hamstrings (for a co-contraction). The orthosis was to be worn always. The client was given specific instructions regarding permissible activity and exercise in the orthosis to reduce the risk of skin sore development. Luke was to continue outpatient therapy once weekly and the client to follow a specific home exercise plan. An

### Box 21.3 Red flag boxes for Case 3

- 1) Would a rehabilitation veterinarian as team leader have made it easier to use other prescribed therapies? Once a veterinarian refers to another, treatment choice is at the discretion of the receiving veterinarian. The surgeon was team leader but had no rehabilitation training
- 2) The individual prescribing and fitting the orthosis is the person to monitor and evaluate orthosis use and complications. This is the person with the most suitable skill set. Some orthotics companies will only work directly with a veterinarian so that follow-up is complete
- 3) The orthosis was reapplied but why had the sores occurred in the first place? Did the orthosis need refitting or changing?
- 4) An orthosis is a medical device; it needs to be fitted correctly and any issues troubleshooted by a therapist with specific training in orthosis fit and application. Regardless, an owner may be non-compliant. Careful, repeated counseling about the risks and importance of proper device use is needed

orthosis recheck evaluation at the rehabilitation facility was set up for 2 weeks later.

Ten days later, 21 weeks following the repair and a few days before the scheduled orthosis recheck and weekly therapy visit, the surgeon called and reported that Luke had jumped off the bed, and was not wearing his orthosis at the time of the incident. Upon examination, it was determined that Luke's Achilles repair had failed. The owner stated that Luke could not wear the orthosis for more than 20 minutes without his skin reddening and due to concerns about a sore, she had been removing it at home. She always took Luke to therapy in the device however (Box 21.3, red flag box 4).

Several options were discussed with the owner: first, repair of the tendon, with the risk of it being a weaker repair and, therefore, more prone to re-injury; second, a pan-tarsal arthrodesis; and third, amputation. The owners elected to perform the arthrodesis.

#### Case 4: Spinal Surgery

A 10-year-old, 45-pound (20kg), spayed female Dalmatian called Marty presented to the veterinary neurologist for acute-onset, rear limb ambulatory paresis. She was observed to have difficulty getting up from sitting and from lying. Also, when ambulating, she placed the dorsum of her right hind paw on the ground or knuckled over on the right hind foot. Marty was taking a non-steroidal anti-inflammatory and a glucosamine chondroitin sulfate supplement for a presumptive diagnosis of osteoarthritis (location unspecified). Physical examination findings included normal pain score, intact segmental spinal reflexes, the presence of deep and superficial sensation, but conscious proprioceptive deficits in the rear limbs (when her paw was placed knuckled over, she did not replace it). Her bladder was "easily expressed."

Further diagnostics were performed. Radiographs showed a T13–L1 and L1–L2 spondylosis, while the myelogram showed extradural compression left T13–L1 and right

L1–L2. Intervertebral disc protrusion was suspected as the cause for the compression and so surgical decompression was recommended. Marty underwent a T13–L2 hemilaminectomy. Her 3-day hospitalization period was without incident and she was discharged on postoperative day 4. On the day of discharge, veterinary examination findings included: absent deep pain sensation in the left hindlimb, hypertonic left hindlimb myotactic (stretch) reflexes, and urinary incontinence with fecal continence. She was discharged with prescriptions for prednisone, phenoxylbenzamine to aid urination, and cephalexin (Box 21.4, red flag box 1). The rehabilitation team was consulted at the time of her discharge. The rehabilitation-certified veterinary technician instructed the clients in the proper administration of the discharge medications, bladder management (including demonstration of techniques for expression), home nursing care (including bedding and preventing sores), making the home safe for Marty, limiting the area she could move in, and dietary changes considering her reduced activity. The physical therapist instructed the clients in basic home rehabilitation techniques including sensory stimulation, passive range of motion (PROM), and patterning of Marty's hindlimbs. The clients were also advised of proper and safe sling-walking techniques, considering the clients' safety with body mechanics. The physical therapist also discussed the potential for use of a cart by Marty as her prognosis for recovery of normal gait was guarded.

Marty returned on postoperative day 13. After recheck and staple removal, she was referred directly for her rehabilitation evaluation. The clients reported that Marty could support weight through her back legs when her bladder was expressed and that they had, since discharge from surgery, restricted her to a small pen. They were using a sling for limited ambulation (outdoors for elimination only). The clients expressed their hope to return Marty to as high a level of function as possible and that they would pursue a cart for ambulation assistance, if necessary.

The physical therapist's evaluation revealed no pain response when palpated. Marty could stand for 30–60 seconds after being placed in that position. She then would slowly “sink down” with her rear limbs into a sit. She had significant postural compensation in standing, including cranial weight shifting onto her forelimbs with her head lowered towards the floor (Box 21.4, red flag box 2).

When walking with assistance, Marty knuckled over onto the dorsum of her rear

paws and adducted her rear limbs with crossing over (Figure 21.6). Her flexor withdrawal response was intact. Deep pain perception was present in the right hind, questionable in the left. Extensor tone was present in both hindlimbs. Conscious proprioception appeared absent in both hindlimbs.

The rehabilitation team, in consultation with the neurologist and clients, set short-term and long-term functional goals. A plan was also outlined and discussed for in-clinic and home therapy. Following this discussion, the clients were given a home program, including sensory stimulation techniques, postural transitions, assisted standing and sling-walking. The clients were instructed in techniques to care for their own backs as well, given Marty's size and dependence upon them for care and mobility.

Marty's rehabilitation progressed as she was seen initially twice weekly for physical rehabilitation. Treatment included: neuromuscular facilitation techniques with functional mobility training and manual therapy (including traction techniques) by the physical therapist, therapeutic exercise (on land and in an aquatic environment), NMES, and LLLT. The home program was progressed to encourage motor learning and carryover from each therapy session. The physical therapist performed all manual therapy and facilitated exercises and performed the initial NMES session while instructing the rehabilitation technician in altering the settings and treatment locations

#### Box 21.4 Red flag boxes for Case 4

- 1) Marty was not discharged on any analgesics; it is important for the rehabilitation team to assess pain before commencing with the rehabilitation plan
- 2) Marty's weight shift forward caused neck and shoulder soreness. Though focused on her return to function, the rehabilitation team could have performed acupuncture and manual therapy techniques earlier in her rehabilitation as a preventative measure. Additionally, the clients should have been instructed not to lift Marty's pelvic girdle too high off the ground as this might have further stressed her
- 3) All neurologic patients should be screened for possible urinary tract infections as bladder emptying can be incomplete due to postural difficulties or neurologic dysfunction of the bladder itself

**Figure 21.6** Rear limbs crossed over. Natural stance of a neurologic Dachshund.



for NMES in subsequent sessions. The rehabilitation technician performed therapeutic exercises, NMES, and LLLT.

Marty was rechecked by the neurologist 4 weeks later. Her prednisone dose was tapered and phenoxybenzamine discontinued as bladder control normalized. Progress included walking with less assistance, needing only support at her left hindlimb. Sit-to-stands now needed only moderate assistance. A new sling and boots to protect the rear feet were provided so that Marty could walk more safely and regularly outdoors (see Chapter 10 for examples).

Full functional re-evaluation was performed 6 weeks after surgery. Marty was occasionally able to be independent during recumbent-to-stand transitions and was independent with recumbent-to-sit transitions. She could walk a few steps independently prior to losing her balance. With harness assistance, she could go further but would adduct her rear limbs or knuckle on the left hind 25–50% of the time. Marty demonstrated good motor control for stand-to-sit transitions >75% of the time. Marty continued to compensate in both sit and standing posture by lowering her head and weight shifting cranially to her forelimbs (Box 21.4, red flag box 2).

Marty continued to progress in her therapy and was seen once weekly for sessions with the physical therapist and rehabilitation veterinary technician. Sessions included facilitation techniques, therapeutic exercise, and LLLT.

At 8 weeks after surgery, Marty could stand for up to 4 minutes independently and recover her standing balance when minimally perturbed. She could walk up to ¼ mile (0.4km) with her assistive harness. Marty regained control of her bladder but full emptying was still ensured with manual expression to prevent infection. The rehabilitation technician reviewed bladder palpation techniques with the clients.

At her 12-week re-evaluation appointment, the clients reported to the veterinary technician that Marty appeared to have some

shoulder and neck soreness (Box 21.4, red flag box 2). They were also concerned that she might have a urinary tract infection, as she was dribbling urine and it was stronger smelling and more concentrated than it was previously (Box 21.4, red flag box 3). The clients had been counseled about signs of infection and so were aware of what to look for. This report was relayed to the physical therapist, as well as to the rehabilitation veterinarian, who recommended a urinalysis and culture. Urine was sampled by cystocentesis aided by the technician and initial analysis by the technician revealed white blood cells and bacteria in the urine. The sample was sent for culture and a broad-spectrum antibiotic prescribed. A non-steroidal anti-inflammatory and acupuncture treatment were both prescribed and administered for the bladder and neck pain. Following veterinary evaluation, the physical therapist applied manual therapy techniques to the neck and shoulder area, localizing and releasing trigger points in muscles surrounding the pectoral girdle. Therapeutic exercises were modified so to decrease stress upon the forelimbs and increase load on the hindlimbs. In addition, fewer repetitions of each exercise (both in clinic and at home) were performed, as it was assumed that multiple repetitions might be causing fatigue of the hindlimb muscles and further stress (through compensation) of the forelimb muscles. The justification for modification of the home program was discussed with the clients by the physical therapist. The veterinary technician instructed them in administration of medications and possible side-effects.

Marty recovered quickly from her urinary tract infection as the correct antibiotic was prescribed for the bacteria cultured. The exercise modifications were found to have reduced neck and shoulder pain. Marty continued to progress in function with periodic re-evaluations by the physical therapist at lengthening intervals. Manual therapies were continued as needed. Marty now walks mostly unassisted but will knuckle over onto the dorsum of her left rear paw periodically

and so wears a protective boot. She continues to be prone to urinary tract infections but her owners are quick to recognize them and seek appropriate treatment.

This case highlights the need for a team approach to all aspects of care, including counseling the owners about potential complications. Luckily, Marty's owners were instructed in signs suspicious for urinary tract infection, including more frequent urination, more concentrated urine, blood in the urine, and/or foul-smelling urine.

### Case 5: Needing a Team

Sam, a 10-year-old male neutered Labrador, presented to Jane Doe who was a certified veterinary technician and had completed a certification course in canine rehabilitation. Sam had just had a TPLO on the right stifle. The rehabilitation facility was housed next to a primary care veterinary clinic. State law read that a technician could work under indirect supervision by a veterinarian, which was interpreted by Jane as meaning a veterinarian could refer directly to her for rehabilitation care and that she would manage the case. Jane owned her own business and was paid by clients directly (Box 21.5, red flag box 1).

Sam was sent to Jane by a local surgeon for routine postoperative rehabilitation. Sam was otherwise healthy. He was started on a program of twice weekly hydrotherapy

in an underwater treadmill followed by balance and proprioceptive work. Home exercises recommended by Jane were flexion and extension range of motion of both stifles and rhythmic stabilization (Box 21.5, red flag box 2).

Sam progressed well until week 6 after surgery, when he developed a grade 4/5 lameness while exercising in the water treadmill. Jane immediately terminated therapy, drained the tank and brought Sam out. After quickly drying him, she examined him closely for signs of heat, swelling, and pain. Jane noted pain on palpation over Sam's cranial thigh muscles on the right. Sam was not painful when his right stifle was extended, but he was painful at end range flexion. He was not painful on pressure medially over the surgical implants.

Jane iced Sam's cranial thigh muscles and his stifle and advised his owner to do the same that evening. Jane's business was not connected to the adjacent primary care clinic, nor was Sam a patient of that clinic. Jane called Sam's surgeon and his primary care veterinarian to facilitate an examination at either the primary care or surgical clinic (Box 21.5, red flag box 3). The primary care veterinarian saw Sam the next day, prescribed a non-steroidal anti-inflammatory (the same one that Sam was previously on for 2 weeks immediately after surgery), and took radiographs of the stifle. The radiographs

#### Box 21.5 Red flag boxes for Case 5

- 1) The term "indirect supervision" is interpreted in the relevant State (province or country) Veterinary Practice Act. In the case of a veterinary technician working in the veterinary medical field under the indirect supervision of a veterinarian, that technician needs to be under the employ of the supervising veterinarian. Liability insurance would be carried by the supervising veterinarian
- 2) Recommending a treatment plan is the same as prescribing a treatment, this can be interpreted as the practice of veterinary medicine
- 3) It is not clear whether the primary care veterinarian or the surgeon is Jane's supervising veterinarian. Jane was right to contact them both, but what if recommendations from each were different?
- 4) Sam's veterinarian(s) who were supervising Jane should have informed her of the plan and outlined their instructions for her in management of the case
- 5) Jane cannot legally make a diagnosis or prescribe a treatment for that diagnosis

were checked by the surgeon and no new pathology was noted. Sam's owners were instructed by the surgeon to confine him to a crate and to rest him for 2 weeks with leash walks to the bathroom only. Jane was informed of the plan for Sam by his owners (Box 21.5, red flag box 4) and despite her concerns, she had no legal power to modify the veterinarian's recommendations. Following the 2 weeks of rest, Sam presented back to Jane's rehabilitation facility. He was no longer on the anti-inflammatories, but exercise was still being restricted to leash walks. On examination, Sam had a grade 3/5 lameness on the right rear which was an improvement since the most recent onset of acute lameness, but prior to that time, Sam's postoperative lameness had been only a grade 2/5. Jane had concerns about persistence of pain. On further examination, Jane found that Sam was painful on direct palpation over his quadriceps muscles on the right and slow stretch of these muscles produced resistance and pain, the left quadriceps had normal stretch with no pain (Figure 21.7).



**Figure 21.7** Quadriceps stretch.

Jane was concerned about a quadriceps strain but had not examined the quadriceps specifically before the acute onset of lameness (Box 21.5, red flag box 5) Jane applied LLLT to the quadriceps and then gently massaged the muscle group. Sam's owners were instructed to massage at home twice daily. Sam visited for laser therapy five times over 2 weeks with improved muscle tone and pain relief noted following each session, but on each return visit the muscle spasm, shortening, and pain had returned. Sam's grade 3/5 lameness persisted.

Jane encouraged Sam's owners to make a recheck appointment with the surgeon so that she could re-evaluate Sam. Sam's treatment notes were sent with the owners so that the surgeon could understand Jane's examination findings. Jane could not write a diagnosis of a quadriceps muscle strain anywhere in the medical record as diagnosis or prescription of a treatment was outside of her scope of practice. Jane was aware that even changing to a therapeutic modality for a new problem was technically illegal without a veterinarian prescribing this therapy. The medical record was carefully worded "pain on palpation over muscles of right cranial thigh, mostly limited to the quadriceps muscle group. Quadriceps stretch shortened compared to the left side. Applied LLLT for symptomatic relief. On return to the rehab center, Sam still had the same pain and shortening of his right quadriceps. It was recommended to have an evaluation by a surgical specialist."

Sam was brought to the surgeon for examination. She confirmed muscle pain but said that it was likely secondary to relative overload from too much activity and recommended a further 2 weeks' rest with leash restriction. A follow-up visit was booked for then. The anti-inflammatory medication prescribed by the primary care veterinarian was resumed for another 2 weeks. Sam's lameness persisted and on repeat examination and radiographs, the surgeon recommended implant removal as his osteotomy was healed but there was a small amount of

bony reaction around the distal screw. The medical notes also indicated some progression of Sam's stifle arthritis.

Following implant removal, Sam was rested for 2 weeks. His lameness appeared to have resolved and so exercise restrictions were lifted by the surgeon. The surgeon recommended returning to Jane for therapy as Sam's right rear leg had "lost muscle." The anti-inflammatory was continued to help manage pain from arthritis. It would be "reduced to lowest effective dose once Sam was back to full fitness."

Sam returned to the rehabilitation center and resumed therapeutic exercises including hydrotherapy, balance work, and cavalettis. Jane measured Sam's thigh circumference and the right was 2 cm less than the left. The home exercise plan was altered by Jane, adding isometric strengthening of the rear limb muscles using front leg lifts (three-legged stands) in addition to his range of motion and rhythmic stabilization exercises.

Sam's first two therapy sessions went well. Jane palpated his quadriceps before and after each therapy session and there was no pain response, although she did not evaluate his response to stretch. Sam returned for the third session with a grade 3/5 right hindlimb lameness. On examination, Jane found pain over the right quadriceps muscle group and shortened stretch, which was painful. Jane treated the muscle spasm and pain with laser therapy and massage and recommended a home heat and stretch exercise. She informed Sam's primary care veterinarian, who recommended leash restriction along with continuing Jane's plan for laser therapy. After six more laser and massage treatments Sam was feeling better and stretch of the right quadriceps had improved. Jane decided that hydrotherapy might be "too much" for Sam and so continued the home exercise plan of front leg lifts progressing to diagonal leg lifts. She added sit-to-stand exercises. Leash walks twice daily were started on the second week of the exercise plan and increased incrementally by 5 minutes each week.

Sam's owners followed the plan and returned for a recheck 4 weeks after the home-only plan was started. Sam appeared comfortable and his thigh circumference measurements were 1 cm different now – an improvement of 1 cm. Jane removed Sam's leash restriction, allowing short periods of time in the back yard.

Sam's owners called to cancel his recheck appointment which had been scheduled for 2 weeks after removal of restrictions. Sam was lame again. He was going to see the orthopedic surgeon. Sam also went to a sports medicine and rehabilitation clinic where he saw a rehabilitation veterinarian and a physical therapist. A grade II quadriceps strain was diagnosed and was treated by the team with therapeutic ultrasound, myofascial trigger point therapy (dry needling), and manual therapies including cross friction massage and mobilization of the coxofemoral and stifle joints. Sam's lameness resolved and he was slowly returned to full activity. His non-steroidal anti-inflammatory medication was discontinued.

### Case 6: Working with Adjunct Professionals

Sophie, a 7-month-old female Irish Wolfhound presented to the Piney Animal Rehabilitation Clinic for an intermittent rear lameness that had suddenly progressed to ambulatory paraparesis in the rear. It was winter and the puppy had been outside on snow and ice with other dogs when the lameness had first been seen.

The owner had initially consulted her chiropractor who was trained in animal chiropractic. The chiropractor had examined the patient and had concerns about working with a neurologic patient with no veterinary diagnosis. A recommendation was made to seek veterinary advice, particularly from a specialist rehabilitation veterinarian (DACVSMR). Following the chiropractor's advice, the owner self-referred her dog to Piney Animal Rehabilitation for initial examination. After the client made the appointment, a technician at the rehabilitation clinic contacted the